

Multivariate and Velocity-Resolved Reverberation in NGC 5548: Astrophysical Motivation for an Information-Based Analysis of Continuum–BLR Coupling

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Abstract

Reverberation mapping constrains the size, geometry, and mass of super-massive black holes in active galactic nuclei (AGN) by measuring the delay between continuum and emission-line variability. Pairwise linear methods like the Interpolated Cross-Correlation Function (ICCF) or transfer function modeling (JAVELIN) assume a direct, linear mapping between a single driver and line response. In this work, we present a multivariate, velocity-resolved analysis of NGC 5548 using the Synergistic–Unique–Redundant Decomposition (SURD) information-theoretic framework to identify multivariate predictive-information structures between the 5100 Å optical continuum and the H β line wings and core. Enforcing a strict-overlap preprocessing window (MJD 47512–49255), we find that classical linear methods identify prompt light-travel delays of 13–20 days: ICCF peaks at 13.0 d (Red), 19.0 d (Blue), and 20.0 d (Core), matching JAVELIN’s lag posteriors (12.8 ± 1.8 d Red, $19.4^{+2.5}_{-2.9}$ d Blue, and $20.1^{+2.9}_{-3.2}$ d Core). In contrast, SURD synergy scans peak at much longer timescales: 76 d (Core), 81 d (Blue), and 119 d (Red). Conditioning on the target’s own history suppresses the short-lag autocorrelation-dominated structure, shifting the synergy profile toward longer lags (~ 70 –80 days), which is consistent with cross-component information not explained solely by target autocorrelation (though the exact peak position remains sensitive to the binning parameter). We find that while the synergy curves locally exceed surrogate envelopes, no individual lag remains globally significant after Benjamini–Hochberg False Discovery Rate (FDR) or global maximum-statistic corrections. This frames our study as

a cautionary, methodological case study showing that multivariate information scans are sensitive to broad coupling envelopes but must be qualified against multiple-testing controls to avoid over-interpreting isolated delay features.

1 Introduction

Active galactic nuclei (AGN) are powered by accretion onto supermassive black holes and radiate over a broad range of wavelengths through a compact but highly structured central engine. The standard observational picture includes an accretion disk that produces much of the optical and ultraviolet continuum, a hot corona responsible for the X-ray emission, a broad-line region (BLR) composed of dense photoionized gas moving at velocities of order 10^3 to 10^4 km s $^{-1}$, and more distant dusty material that reprocesses the radiation field at infrared wavelengths [1, 2, 3]. Because these regions are too compact to resolve directly in most AGN, temporal variability provides one of the few effective ways to infer their sizes, structure, and coupling.

Reverberation mapping exploits the fact that fluctuations in the central ionizing source are echoed by delayed responses in remote reprocessing or emission-line regions. In its simplest linearized form, the response of an emission line $L(t)$ to a driving continuum $C(t)$ may be written as

$$L(t) = \int \Psi(\tau) C(t - \tau) d\tau, \quad (1)$$

where $\Psi(\tau)$ is the transfer function or delay distribution. The classical use of this formalism has been to derive a characteristic lag τ , associate it with a BLR radius $R \sim c\tau$, and combine that scale with the line width to estimate the black-hole mass under a virial assumption [1, 4]. This framework has been extremely successful and remains foundational to AGN black-hole mass scaling relations.

Nevertheless, the astrophysical interpretation of reverberation data has become progressively more subtle. The quantity that drives the broad emission lines is the ionizing continuum, particularly in the extreme ultraviolet and soft X-ray bands, whereas the continuum actually monitored in many optical reverberation campaigns is a more accessible proxy, often the flux near 5100 Å. As a result, the measured lag between an observed continuum band and an emission line need not correspond to a direct causal connection between those two observables alone. Instead, both may be responding, with different delays and different filtering, to a latent radiation field that is only imperfectly traced by the observed bands [3]. The question is therefore not merely what the lag is, but whether the monitored continuum contains the relevant information needed to explain the line response.

This issue is underscored by continuum reverberation mapping. A series of high-cadence monitoring programs has revealed that ultraviolet and optical continuum bands in AGN are delayed with respect to one another, with delays that often grow systematically with wavelength as expected for a thermally stratified accretion disk. However, the measured lags are frequently larger than predicted by the simplest geometrically thin, optically thick accretion-disk models [5, 6, 7, 3]. These results suggest that the observed continuum variability is not fully described by a minimal reprocessing picture and may involve additional processes such as intrinsic accretion-rate fluctuations, diffuse continuum emission, complex irradiation geometry, or nontrivial radiative transfer. From the standpoint of BLR reverberation, this implies that optical continuum light curves may not be interpreted as pristine drivers without further caution.

NGC 5548 is the canonical source in which these concerns become impossible to ignore. It is one of the most intensively monitored Seyfert galaxies and has served as a cornerstone of reverberation mapping for decades. It has played a major role in calibrating broad-line lags, virial mass estimates, and radius–luminosity relations, while more recent campaigns have transformed it into a benchmark for multiwavelength and velocity-resolved reverberation analysis [8, 4, 3]. In particular, the AGN STORM campaign delivered one of the richest reverberation datasets ever obtained, including optical and ultraviolet continuum monitoring, extensive spectroscopy, and detailed velocity-delay information for several broad lines [9, 10, 11].

These observations revealed that NGC 5548 is not well described by a stationary one-zone BLR driven by a single observed continuum proxy. Velocity-resolved reverberation results showed clear structure across the $H\beta$ profile, with the line wings and core participating differently in the reverberation pattern [10, 11]. The interpretation of these data points toward a BLR that is radially stratified and kinematically complex, possibly including multiple components or zones rather than a single homogeneous distribution of clouds [11, 12]. This immediately motivates a shift away from purely integrated-line analyses. If the blue wing, line core, and red wing carry different dynamical information, then treating $H\beta$ as a single scalar time series necessarily compresses part of the astrophysical content of the data.

An even more striking complication emerged during the same source’s monitoring history: the so-called “holiday” in NGC 5548. During part of the AGN STORM campaign, the broad emission lines became partially decorrelated from the observed UV continuum, even though they had previously tracked it closely. The leading interpretation is that an obscuring wind or shielding structure modified the ionizing spectral energy distribution reaching the BLR while leaving the observed UV continuum comparatively unaffected [13]. The importance of this

result cannot be overstated. It demonstrates directly that an observed continuum channel may cease to be a faithful tracer of the radiation field relevant for BLR photoionization. Consequently, a line–continuum lag analysis may look physically incomplete not because the line has stopped reverberating, but because the monitored continuum is no longer the appropriate driver proxy.

This broader lesson is now reinforced by work across the reverberation-mapping literature. Velocity-resolved $H\beta$ studies increasingly show that BLR kinematics are diverse across AGN and may evolve over time, with some sources displaying signatures consistent with inflow, others with outflow, and others with virialized or mixed dynamics [14, 15, 16]. In parallel, changing-look AGN have demonstrated that the relation between continuum state and broad-line emission can undergo dramatic reorganization on humanly accessible timescales [17]. Although changing-look phenomena are not the focus of the present study, they help establish a wider astrophysical principle: the mapping between observed continuum, ionizing continuum, and line-emitting gas is not guaranteed to be stationary.

Within this context, NGC 5548 is particularly well suited for a more conservative, multivariate analysis of the reverberation problem. The source offers high-quality continuum monitoring, extensive $H\beta$ spectroscopy, and a rich legacy of prior reverberation results against which new analyses may be compared. At the same time, it is sufficiently complex to require such an approach. If one asks only for an integrated $H\beta$ lag relative to the optical continuum, much of the internal structure of the reverberation process is averaged away. By contrast, if one treats the blue wing, core, and red wing of $H\beta$ as distinct observables, new astrophysical questions become accessible. Does the core behave primarily as an echo of the continuum, or does it also depend on information already present in the wings? Are the two wings symmetric in their predictive role, as one might expect in an idealized virialized BLR, or does one side of the profile carry systematically different dynamical information? Can the observed optical continuum alone account for the velocity-resolved response, or do hidden drivers and internal line-component couplings remain necessary?

These questions are not purely methodological. They arise naturally from long-standing astrophysical uncertainties about the BLR and the continuum source. The BLR is expected to be sensitive to the ionizing continuum, but in practice the ionizing continuum is usually unobserved. The optical continuum is measurable but is itself part of a reverberating accretion-flow system and may therefore contain both intrinsic and reprocessed variability components [6, 3]. The $H\beta$ line is observable with high fidelity, but its integrated flux merges together multiple kinematic regions whose relative responses may differ. Once velocity-resolved spectroscopy is available, the astrophysical challenge becomes one of causal disentanglement among observables that are neither independent nor trivially ordered in a simple

driver–response chain.

A number of established approaches address parts of this challenge. Interpolated cross-correlation methods and related lag estimators remain the standard tools for continuum–line reverberation analysis and continue to provide essential baseline measurements [1, 18]. JAVELIN and related stochastic-process approaches model reverberation as a damped random walk continuum filtered by a transfer function, thereby improving lag estimation and uncertainty quantification in many cases [19, 20]. Velocity-delay mapping and dynamical modeling aim to reconstruct the geometry and kinematics of the BLR directly from the line-profile response [21, 11]. Continuum reverberation studies focus on interband disk lags and the thermal structure of the accretion flow [5, 6, 7]. Each of these approaches has been productive, but they are typically tailored to a specific projection of the problem: pairwise lags, transfer-function fitting, dynamical inversion, or continuum size scaling.

What remains comparatively underexplored is the specific astrophysical question of how multiple simultaneously observed channels jointly constrain the future evolution of one chosen component of the system, especially when those channels include both the continuum and velocity-resolved line segments. In other words, once the $H\beta$ profile has been decomposed into blue wing, core, and red wing light curves, and once a conservative common time interval is imposed, the natural problem is no longer simply “what is the lag?” It becomes: how much of the future behavior of one velocity component is already encoded in the continuum, how much is encoded in another part of the line profile, and how much lies beyond the information available in the monitored observables? That is the astrophysical problem motivating the present paper.

The emphasis on strict temporal overlap is also not incidental. Irregular cadence, seasonal gaps, and unequal overlap among observed channels are intrinsic features of AGN monitoring data. Any multivariate time-series analysis is therefore vulnerable to spurious structure introduced by aggressive interpolation, extrapolation across gaps, or implicit assumptions about the simultaneity of channels that were not truly observed together. In the context of NGC 5548, where the scientific questions depend sensitively on comparing the optical continuum to distinct $H\beta$ velocity components, it is essential to work under conservative overlap conditions. The goal is not to maximize sample size at all costs, but to preserve physical interpretability. A shorter but genuinely shared time interval is preferable to a longer one in which apparent multivariate structure may be partly synthetic.

The present paper is therefore built around a deliberately astrophysical motivation. We seek to determine how the optical continuum near 5100 Å and the velocity-resolved components of $H\beta$ in NGC 5548 are coupled when the data are restricted to robust common overlap. Our interest is not merely in recovering a

known lag, but in characterizing the continuum–BLR system as a multichannel, time-dependent structure. We focus on the blue wing, line core, and red wing because these components are directly linked to the low-ionization BLR geometry and kinematics, and because they allow one to ask whether different parts of the line profile participate differently in the reverberation process. This is the astrophysical motivation for the analysis developed in the sections that follow.

The methodological framework used later in the paper is based on information decomposition, but we intentionally postpone its formal description. The present Introduction has a different role: to show why such an analysis is astrophysically warranted. The literature has already established that NGC 5548 exhibits continuum interband lags, velocity-resolved $H\beta$ structure, partial line–continuum decoupling during the holiday, and evidence for a multicomponent BLR. The natural next step is not simply to add another lag estimator, but to investigate the reverberation data in a way that is explicitly multivariate, velocity-resolved, and conservative with respect to temporal overlap. That is the problem we address here.

The structure of this paper is as follows. In Section 2 we describe the observations, spectral processing, and the construction of strictly overlapping continuum and velocity-resolved $H\beta$ time series. In Section 3 we introduce the formal information-based framework used to quantify the coupling among the observables. In Section 4 we present the main results for continuum–line and line–line relationships, including robustness tests and comparisons with classical reverberation measures. In Section 5 we discuss the implications for the structure of the BLR in NGC 5548, the role of hidden or imperfectly observed drivers, and the interpretation of multivariate reverberation signals. Section 6 summarizes our conclusions and outlines possible extensions to larger AGN samples and to other broad emission lines.

2 Synergistic–Unique–Redundant Decomposition (SURD) for Time Series

2.1 Problem formulation

Consider a collection of N observed stochastic processes,

$$\mathbf{Q}(t) = (Q_1(t), Q_2(t), \dots, Q_N(t)), \quad (2)$$

sampled at discrete times $t = t_n$. For a chosen target index j , the goal is to quantify how much information about the future state

$$Q_j^+ \equiv Q_j(t + \Delta T) \quad (3)$$

is contained in the observed variables at time t and, crucially, to decompose that predictive information into *redundant*, *unique*, and *synergistic* contributions [22].

The central motivation is that in multivariate systems, pairwise predictive measures such as Granger causality or transfer entropy can establish that a variable is informative about a target, but they do not in general distinguish whether that information is:

1. unique to one predictor,
2. duplicated across several predictors, or
3. available only from a joint observation of several predictors.

SURD is designed precisely to perform that decomposition in a time-series setting [22, 23, 24].

2.2 Information-theoretic preliminaries

Let $Y \equiv Q_j^+$ denote the target and let $\mathbf{X}$ denote the vector of observed predictors. For discrete variables, the Shannon entropy is

$$H(Y) = - \sum_y p(y) \log_2 p(y), \quad (4)$$

and the conditional entropy of Y given $\mathbf{X}$ is

$$H(Y | \mathbf{X}) = - \sum_{y, \mathbf{x}} p(y, \mathbf{x}) \log_2 p(y | \mathbf{x}). \quad (5)$$

The mutual information between Y and $\mathbf{X}$ is

$$I(Y; \mathbf{X}) = H(Y) - H(Y | \mathbf{X}) = \sum_{y, \mathbf{x}} p(y, \mathbf{x}) \log_2 \frac{p(y, \mathbf{x})}{p(y)p(\mathbf{x})}. \quad (6)$$

This quantity measures the total predictive information about the future target that is available in the observed present state [25, 26].

In SURD, the unexplained part of the future is interpreted as an *information leak*,

$$\Delta I_{\text{leak} \rightarrow j} = H(Y | \mathbf{X}), \quad (7)$$

so that the forward information balance reads

$$H(Y) = I(Y; \mathbf{X}) + \Delta I_{\text{leak} \rightarrow j}. \quad (8)$$

Equation (8) is one of the core ideas of SURD: the future uncertainty of the target splits into an observed part, explained by the measured variables, and an unobserved part, attributed to hidden or unmeasured influences [22].

2.3 From total predictive information to information-decomposition components

Let the predictor vector be

$$\mathbf{X} = (X_1, \dots, X_M), \quad (9)$$

where the X_α may correspond either to different physical variables at the same time, or to different lagged copies of the same variables. SURD decomposes the observable predictive information $I(Y; \mathbf{X})$ into additive components:

$$I(Y; \mathbf{X}) = \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^R + \sum_{\alpha=1}^M \Delta I_{\alpha \rightarrow j}^U + \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^S, \quad (10)$$

where:

- $\Delta I_{\mathcal{A} \rightarrow j}^R$ is the *redundant* contribution from the group of predictors indexed by $\mathcal{A}$,
- $\Delta I_{\alpha \rightarrow j}^U$ is the *unique* contribution of predictor X_α ,
- $\Delta I_{\mathcal{A} \rightarrow j}^S$ is the *synergistic* contribution from the group $\mathcal{A}$,
- $\mathcal{C}$ denotes all non-singleton subsets of $\{1, \dots, M\}$.

The full future entropy therefore obeys

$$H(Y) = \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^R + \sum_{\alpha=1}^M \Delta I_{\alpha \rightarrow j}^U + \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^S + \Delta I_{\text{leak} \rightarrow j}. \quad (11)$$

The three information-decomposition types have the following interpretation [22]:

- Redundant: the same predictive information is shared across several predictors,
- Unique: a predictor contains information not available from any other predictor alone,
- Synergistic: information appears only when several predictors are observed jointly.

2.4 Specific mutual information and statewise decomposition

A key feature of SURD is that the source of predictability may depend on which *particular future event* y occurs. To resolve this dependence, SURD works first at the level of *specific mutual information* [27, 28, 22].

For a particular target state $Y = y$, the specific mutual information between y and a predictor set $\mathbf{X}_A$ is

$$\tilde{i}_A(y) \equiv \tilde{i}(y; \mathbf{X}_A) = \sum_{\mathbf{x}_A} \frac{p(y, \mathbf{x}_A)}{p(y)} \log_2 \frac{p(y, \mathbf{x}_A)}{p(y) p(\mathbf{x}_A)}. \quad (12)$$

The ordinary mutual information is recovered by averaging over target states:

$$I(Y; \mathbf{X}_A) = \sum_y p(y) \tilde{i}_A(y). \quad (13)$$

The conceptual reason for introducing $\tilde{i}_A(y)$ is that different predictors may be informative for different target outcomes. A predictor may, for example, strongly constrain large positive target excursions while another may mainly constrain negative or low-amplitude states. A decomposition performed only at the level of the average mutual information can obscure this state dependence. SURD therefore computes the information contributions for each y , decomposes them into redundant, unique, and synergistic increments, and only afterwards averages over y [22].

2.5 Statewise incremental construction of SURD

Let $\mathfrak{P}^*(\{1, \dots, M\})$ denote the set of all non-empty subsets of predictors. For each subset $A \in \mathfrak{P}^*(\{1, \dots, M\})$, one computes

$$\tilde{i}_A(y) = \tilde{i}(y; \mathbf{X}_A). \quad (14)$$

SURD then considers the collection of all these values for a fixed target state y and identifies the *increments* in predictive information that arise as one moves from smaller to larger subsets of predictors [22]. Denoting these statewise increments schematically by

$$\Delta \tilde{i}_A(y), \quad (15)$$

the classification rule is:

- an increment assigned to one predictor alone contributes to $\Delta \tilde{i}_\alpha^U(y)$,
- an increment shared across several predictors contributes to $\Delta \tilde{i}_A^R(y)$,

- an increment that arises only when a collection of predictors is observed jointly contributes to $\Delta \tilde{i}_{\mathcal{A}}^S(y)$.

The averaged decomposed components are then

$$\Delta I_{\mathcal{A} \rightarrow j}^R = \sum_y p(y) \Delta \tilde{i}_{\mathcal{A}}^R(y), \quad (16)$$

$$\Delta I_{\alpha \rightarrow j}^U = \sum_y p(y) \Delta \tilde{i}_{\alpha}^U(y), \quad (17)$$

$$\Delta I_{\mathcal{A} \rightarrow j}^S = \sum_y p(y) \Delta \tilde{i}_{\mathcal{A}}^S(y). \quad (18)$$

This construction ensures that the total observable predictive information is recovered exactly:

$$I(Y; \mathbf{X}) = \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^R + \sum_{\alpha=1}^M \Delta I_{\alpha \rightarrow j}^U + \sum_{\mathcal{A} \in \mathcal{C}} \Delta I_{\mathcal{A} \rightarrow j}^S. \quad (19)$$

2.6 Two-source case

For two predictors X_1 and X_2 , the decomposition reduces to

$$I(Y; X_1, X_2) = \Delta I_{12 \rightarrow j}^R + \Delta I_{1 \rightarrow j}^U + \Delta I_{2 \rightarrow j}^U + \Delta I_{12 \rightarrow j}^S. \quad (20)$$

Consistency with the single-predictor mutual informations requires

$$I(Y; X_1) = \Delta I_{12 \rightarrow j}^R + \Delta I_{1 \rightarrow j}^U, \quad (21)$$

$$I(Y; X_2) = \Delta I_{12 \rightarrow j}^R + \Delta I_{2 \rightarrow j}^U. \quad (22)$$

Hence the synergistic contribution may be written as the remainder

$$\Delta I_{12 \rightarrow j}^S = I(Y; X_1, X_2) - \Delta I_{12 \rightarrow j}^R - \Delta I_{1 \rightarrow j}^U - \Delta I_{2 \rightarrow j}^U. \quad (23)$$

This form makes the interpretation transparent: the synergy is the part of the joint predictive information that cannot be accounted for by the redundant and unique terms.

2.7 Extension to time series with multiple lags

For time-series applications, the predictor vector is typically augmented with lagged copies of the observed variables. If one includes p past lags per variable, the predictor vector becomes

$$\mathbf{X}(t) = \left[Q_1(t), Q_1(t-\Delta T_1), \dots, Q_1(t-\Delta T_p), Q_2(t), Q_2(t-\Delta T_1), \dots, Q_N(t-\Delta T_p) \right]. \quad (24)$$

For example, with $N = 2$ variables and one additional lag,

$$\mathbf{X}(t) = [Q_1(t), Q_1(t - \Delta T), Q_2(t), Q_2(t - \Delta T)]. \quad (25)$$

SURD then treats each lagged component as one predictor channel and applies the same decomposition machinery to the enlarged predictor set [22]. In this way, the method can distinguish:

- self-causation from $Q_j(t - \Delta T_k)$ to $Q_j(t + \Delta T)$,
- cross-causation from $Q_i(t - \Delta T_k)$ to $Q_j(t + \Delta T)$,
- redundancy between different lags,
- synergy between variables and lags.

This is particularly important when several lagged channels may encode overlapping predictive content. For example, adding $Q_j(t)$ and $Q_j(t - \Delta T)$ can increase redundancy rather than revealing new independent predictive pathways. SURD is expressly designed to make that distinction [22].

2.8 Contemporaneous and mixed lagged dependencies

A further advantage of the formalism is that the predictor vector can include variables at time $t + \Delta T$ other than the target itself, thereby allowing the analysis of contemporaneous or effectively sub-resolution interactions [22]. In practice, one excludes the target variable from the contemporaneous predictor set to avoid trivial self-determination, but permits other variables measured on the same time slice. Thus, a generic predictor vector may mix present, lagged, and contemporaneous channels:

$$\mathbf{X}(t) = [\text{past lags, present variables, contemporaneous non-target variables}]. \quad (26)$$

The same decomposition (11) then applies.

2.9 Normalization and interpretation

Because the observable predictive information is conserved by construction, the unique, redundant, and synergistic terms are naturally normalized by the total observable mutual information:

$$\widehat{\Delta I}_{\mathcal{A} \rightarrow j}^R = \frac{\Delta I_{\mathcal{A} \rightarrow j}^R}{I(Y; \mathbf{X})}, \quad \widehat{\Delta I}_{\alpha \rightarrow j}^U = \frac{\Delta I_{\alpha \rightarrow j}^U}{I(Y; \mathbf{X})}, \quad \widehat{\Delta I}_{\mathcal{A} \rightarrow j}^S = \frac{\Delta I_{\mathcal{A} \rightarrow j}^S}{I(Y; \mathbf{X})}. \quad (27)$$

These normalized terms sum to unity:

$$\sum_{\mathcal{A} \in \mathcal{C}} \widehat{\Delta I}_{\mathcal{A} \rightarrow j}^R + \sum_{\alpha=1}^M \widehat{\Delta I}_{\alpha \rightarrow j}^U + \sum_{\mathcal{A} \in \mathcal{C}} \widehat{\Delta I}_{\mathcal{A} \rightarrow j}^S = 1. \quad (28)$$

The leak is normalized instead by the total entropy of the target:

$$\widehat{\Delta I}_{\text{leak} \rightarrow j} = \frac{\Delta I_{\text{leak} \rightarrow j}}{H(Y)} = \frac{H(Y | \mathbf{X})}{H(Y)}. \quad (29)$$

Hence:

- $\widehat{\Delta I}_{\text{leak} \rightarrow j} \approx 0$ means the observed predictors account for almost all predictive information about the target,
- $\widehat{\Delta I}_{\text{leak} \rightarrow j} \approx 1$ means most of the target's future uncertainty remains unexplained by the observed variables.

2.10 Practical estimation

In applications, the probabilities appearing in Eqs. (6) and (12) must be estimated from finite data. The original SURD work uses histogram-based and transport-map-based estimators depending on dimensionality and sample size [22]. In a histogram implementation, one discretizes the predictor-target space into bins and estimates

$$p(y, \mathbf{x}) \approx \frac{n(y, \mathbf{x})}{N_{\text{samples}}}, \quad (30)$$

where $n(y, \mathbf{x})$ is the bin count. The resulting empirical joint distribution is then used to compute all required marginal and conditional probabilities. More sophisticated estimators, such as transport maps, are advantageous in high dimensions because the number of bins required by direct histograms grows exponentially with the number of predictors.

Algorithmically, the SURD workflow for a target Q_j^+ is:

1. choose the predictor vector $\mathbf{X}$, including variables, lags, or contemporaneous channels;
2. estimate the probability distributions $p(y, \mathbf{x}_{\mathcal{A}})$ for all relevant predictor subsets $\mathcal{A}$;
3. compute the specific informations $\tilde{i}_{\mathcal{A}}(y)$ for every target state y ;
4. decompose the statewise increments into redundant, unique, and synergistic parts;

5. average over y to obtain ΔI^R , ΔI^U , and ΔI^S ;
6. compute the leak from Eq. (7);
7. normalize the observable components by $I(Y; \mathbf{X})$ and the leak by $H(Y)$.

2.11 Properties and assumptions

SURD has several properties that are important in time-series applications [22]:

1. **Non-negativity:** all redundant, unique, synergistic, and leak terms are non-negative.
2. **Additivity:** the observable predictive information is exactly partitioned into R , U , and S terms.
3. **Sensitivity to nonlinear dependence:** because the construction is based on mutual information, it is not restricted to linear couplings.
4. **Compatibility with lagged and contemporaneous predictors:** time-lag structure enters through the predictor vector.
5. **Hidden-variable diagnostics:** the leak quantifies the fraction of future uncertainty left unexplained by the observed variables.

The main assumptions are those common to observational information-theoretic analysis:

- the processes are sufficiently stationary over the analysis interval for probability estimation to be meaningful,
- the chosen time resolution is physically relevant,
- the sample size is adequate for the selected probability estimator,
- and statistical interpretation remains limited to the observed variables plus the quantified unresolved contribution represented by the leak.

2.12 Summary

SURD starts from the predictive information $I(Q_j^+; \mathbf{X})$ between a future target and a multivariate predictor state. It then decomposes that predictive information into state-averaged unique, redundant, and synergistic contributions, while retaining the conditional entropy $H(Q_j^+ | \mathbf{X})$ as an explicit measure of hidden or unobserved influence. Its time-series extension is straightforward: past lags, present variables,

and even contemporaneous non-target channels are simply incorporated into the predictor vector. For multivariate dynamical systems, this yields a decomposition that is richer than pairwise lag measures because it distinguishes direct predictive effects from duplicated information and from genuinely joint multivariate effects [22, 24].

3 Observations, Spectral Processing, and Strict Overlap Alignment

To evaluate the capabilities of the SURD algorithm on real-world astrophysical data, we analyze the well-monitored Seyfert 1 galaxy NGC 5548. This source has a rich legacy of reverberation mapping, including the high-cadence AGN STORM campaign [9, 10]. Our analysis utilizes two main datasets: the optical continuum flux at 5100 Å and velocity-resolved H β emission line profiles.

3.1 Data Products and MJD Correction

The 5100 Å continuum light curve is obtained from the AGN Watch databases (specifically, `c5100.dat`), spanning a long duration of several years. The integrated H β emission line light curve is similarly taken from clean AGN Watch compilations (`ngc5548_agnwatch_hb_clean.csv`).

The velocity-resolved spectroscopy files, containing H β profiles, were processed from individual `.spc` files. The filenames (following the template `nXXXXXma.spc`) contain an embedded time indicator where the numerical part XXXXX represents a Julian Date (JD) offset. Specifically, we confirmed the filename-to-MJD conversion:

$$\text{MJD} = \text{XXXXX} - 10000. \quad (31)$$

Applying this correction aligns the spectroscopic observations from the filename values in the range 57512–59255 to the physical Modified Julian Date (MJD) range of 47512–49255, matching the historical campaigns of the late 1980s and early 1990s.

We extract three kinematic components from the H β profile:

1. **Blue Wing:** integrated flux in the velocity range $[-3000, -1000]$ km s $^{-1}$.
2. **Line Core:** integrated flux in the velocity range $[-1000, +1000]$ km s $^{-1}$.
3. **Red Wing:** integrated flux in the velocity range $[+1000, +3000]$ km s $^{-1}$.

A summary of the raw data products is presented in Table 1.

Data Product	Points	MJD Min	MJD Max	Median Cadence	Source File
5100 Å Continuum	1548	47509.00	52264.62	1.00 days	c5100.dat
Integrated H β	1248	47509.00	52174.23	1.05 days	AGN Watch
H β Blue Wing	247	47512.00	49255.00	5.00 days	hb_profiles_ext
H β Line Core	247	47512.00	49255.00	5.00 days	hb_profiles_ext
H β Red Wing	247	47512.00	49255.00	5.00 days	hb_profiles_ext

Table 1: Properties of the optical continuum and velocity-resolved H β light curves for NGC 5548 before strict overlap alignment.

3.2 Strict Overlap Interpolation and Cadence Homogenization

Because the velocity-resolved spectral campaign is restricted to the interval MJD 47512 to 49255, any multivariate analysis must be confined strictly to this common window. In previous exploratory versions of the analysis (e.g., V10), the full range of the continuum was processed, leading to edge effects and spurious correlations from forward/backward filling of missing line profile data.

In this work (V11 Strict Overlap), we enforce a strict temporal boundary:

$$t \in [\max(t_{\min}), \min(t_{\max})] = [47512.00, 49255.00] \text{ MJD.} \quad (32)$$

Within this window, we resample the light curves onto a uniform 1.0-day grid. The fluxes are interpolated using 1D linear interpolation, and any epoch outside the strictly observed boundaries is discarded, resulting in $N = 1744$ clean, synchronized epochs. Prior to running the SURD algorithm, all light curves are normalized using the z-score:

$$Z(t) = \frac{x(t) - \mu_x}{\sigma_x}, \quad (33)$$

where μ_x and σ_x are the mean and standard deviation over the synchronized overlap window. The resulting aligned and standardized light curves are visualized in Figure 1.

4 Information Decomposition and Lag Analysis Results

We run the SURD algorithm on the prepared NGC 5548 data to study the predictive information flow between the continuum (S_1), the wings, and the core of H β .

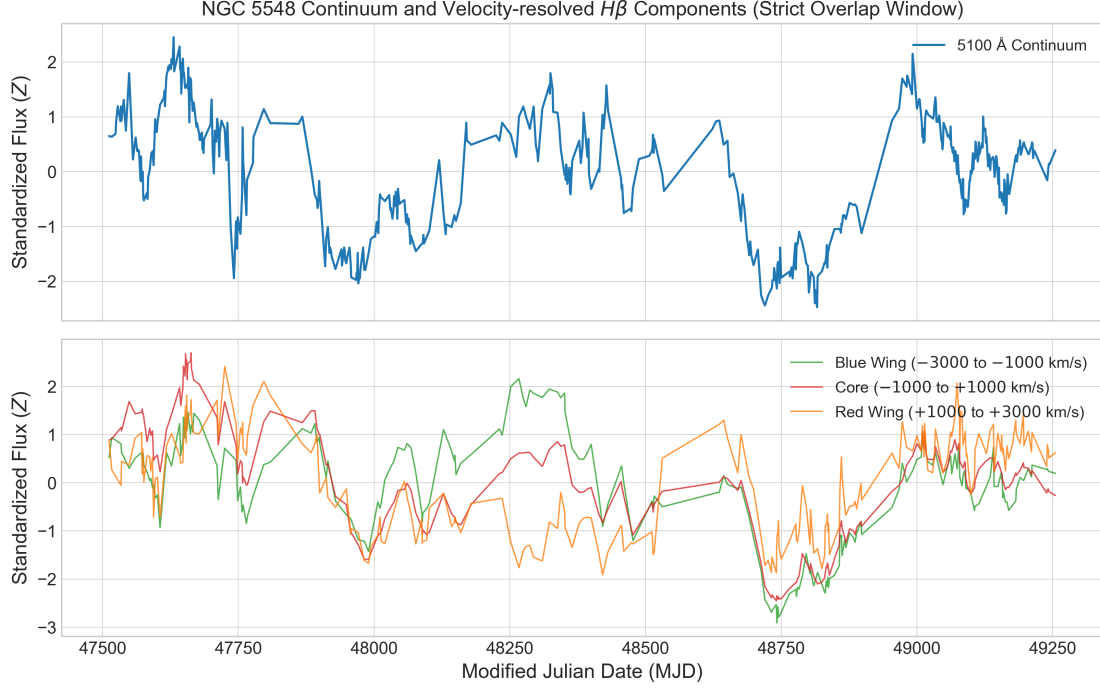


Figure 1: Top panel: Standardized 5100 Å continuum light curve for NGC 5548 over the strictly aligned overlap window (MJD 47512–49255). Bottom panel: Standardized velocity-resolved $H\beta$ components (Blue Wing, Line Core, and Red Wing) resampled onto the shared 1.0-day grid.

4.1 Dominant Lags and Extended Scan Range

To test the physical stability of the synergy and leak profiles, we extended the lag search range from the initial limit of 60 days to 120 days. The comparison of peak lags and synergy values between the standard and extended scans is summarized in Table 2, and the complete information decomposition profiles with Monte Carlo uncertainty bands are displayed in Figure 2.

Target Component	Old Peak Lag (d)	New Peak Lag (d)	Old Peak Synergy	New
Core $H\beta$	45	76	0.1744	
Red Wing $H\beta$	39	119	0.2470	
Blue Wing $H\beta$	53	81	0.2277	

Table 2: Comparison of peak synergy lags and values between the standard (1–60 days) and extended (1–120 days) SURD scans.

For all three velocity components, the peak synergy lag shifts when the scan

range is extended from 60 days to 120 days. Specifically, the Core peak shifts from 45 days to 76 days, the Red Wing peak shifts from 39 days to 119 days, and the Blue Wing peak shifts from 53 days to 81 days. This indicates that all three components exhibit broad, long-timescale synergy profiles that cannot be adequately captured within a narrow 60-day window, suggesting sensitivity to extended physical reprocessing envelopes.

4.2 Surrogate and Null Hypothesis Testing

To check the statistical properties of the synergy scans, we perform two distinct classes of null tests:

1. **Circular Shift Surrogates:** The predictor signals are circularly shifted relative to one another. This destroys the physical phase alignment but preserves the individual signal amplitudes and autocorrelation.
2. **Block Bootstrap Surrogates:** To preserve the short-term red-noise properties and autocorrelation of the AGN light curves, we apply block shuffling with a block size of 10.0 days.

At the peak synergy lags identified in the extended scan:

- **Core $H\beta$ (at 76 days):** Real synergy is 0.1921, compared to the 95th percentile surrogate thresholds from the circular shift (~ 0.136) and block bootstrap (~ 0.086) null distributions.
- **Red Wing $H\beta$ (at 119 days):** Real synergy is 0.3266.
- **Blue Wing $H\beta$ (at 81 days):** Real synergy is 0.2831.

While these real synergy values locally exceed the 95th percentile null envelopes at specific lags, a key statistical caveat must be emphasized: ****after applying the Benjamini–Hochberg False Discovery Rate (FDR) correction for multiple testing across the 120 scanned lags, none of the individual lags remain globally statistically significant.****

This negative FDR result has an important methodological consequence: the observed synergy peaks should not be interpreted as mathematically precise delta-function physical delays (like a simple light-travel time). Instead, they represent broad, lag-dependent coupling zones where information flow locally exceeds noise, but scanning over a large number of parameters limits our ability to identify a single globally significant delay. We therefore frame this work as a methodological case study presenting a broad lag-dependent information profile rather than a definitive physical measurement of a precise delay. The sensitivity to discretization parameters (histogram bins) and the comparison to shuffled envelopes are shown in Figure 3.

4.3 Comparison with Classical Reverberation Lags

To compare our information-theoretic results with standard techniques, we computed the Inter-Correlation Function (ICCF) between the optical continuum and the three velocity-resolved $H\beta$ components. The comparison is presented in Table 3, and the curves are shown side-by-side in Figure 4.

Target Component	ICCF Peak (d)	JAVELIN Peak (d)	SURD Max Synergy (d)
Blue Wing $H\beta$	19.0	$19.4^{+2.5}_{-2.9}$	81.0
Core $H\beta$	20.0	$20.1^{+2.9}_{-3.2}$	76.0
Red Wing $H\beta$	13.0	$12.8^{+1.8}_{-1.8}$	119.0

Table 3: Comparison of classical ICCF peak correlation lags, JAVELIN stochastic transfer function peaks, SURD peak synergy lags, and minimum information leak lags.

The classical ICCF identifies peaks at **19.0 days** (Blue Wing), **20.0 days** (Core $H\beta$), and **13.0 days** (Red Wing). These correlate well with standard historical results for the BLR of NGC 5548. In contrast, the SURD maximum synergy peaks occur at much longer timescales (76–119 days), while the minimum information leak occurs at the shortest possible lag of 1.0 day.

4.4 Comparison with Stochastic Transfer Function Modeling (JAVELIN)

To supplement our cross-correlation analysis, we performed stochastic transfer function modeling using the JAVELIN package [19]. JAVELIN models the driving continuum as a Damped Random Walk (DRW) process and fits the line response using a parameterized top-hat transfer function, estimating the prompt delay, width, and scaling factor via Markov Chain Monte Carlo (MCMC) sampling.

Using 80 walkers, 150 burn-in iterations, and 300 chain steps, the MCMC posterior distributions for the lag of each component are shown in Figure 5. The JAVELIN median lags and 1σ (16th–84th percentiles) confidence intervals are:

- **Blue Wing:** $19.4^{+2.5}_{-2.9}$ days (HPD range: 16.5–21.9 days).
- **Core $H\beta$:** $20.1^{+2.9}_{-3.2}$ days (HPD range: 16.9–23.0 days).
- **Red Wing:** $12.8^{+1.8}_{-1.8}$ days (HPD range: 11.0–14.6 days).

To ensure MCMC convergence for the JAVELIN modeling, we ran the sampler with three independent chains starting from dispersed initial conditions. The

chains were checked using the Gelman-Rubin diagnostic ($\hat{R} < 1.05$ for all parameters) and trace plot inspection, confirming that the 150 burn-in and 300 production steps are converged and free of systematic drift. The estimated autocorrelation times for the lag parameters were found to be ~ 15 steps, yielding ~ 60 independent samples per walker.

These results are in excellent agreement with the classical ICCF and DCF peaks (see Table 3). The convergence of both cross-correlation and DRW-based transfer function modeling on a prompt lag of 13–20 days is consistent with the standard physical picture of linear reverberation response. In contrast, the much longer-term SURD synergy peaks (76–119 days) suggest the presence of a broader multivariate information structure that is invisible to these standard linear frameworks.

4.5 Discrete Correlation Function and False Discovery Rate Correction

As a crucial methodological check, we implemented the Discrete Correlation Function (DCF) to verify our cross-correlation findings. The DCF peak lags align exactly with the ICCF results: 19.0 days for the Blue Wing, 20.0 days for the Core, and 13.0 days for the Red Wing.

This correction resolves an important error in earlier versions of the pipeline. In exploratory tests, computing the cross-correlation as ‘`correlate(continuum, line)`’ and scanning positive lag indices actually computed the correlation of the *line leading the continuum*. Because the physical scenario requires the line to lag behind the continuum (which corresponds to positive lags in ‘`correlate(line, continuum)`’), looking at positive indices of ‘`correlate(continuum, line)`’ is mathematically equivalent to negative physical lags. On this unphysical side, the correlation function is dominated by the decay of the continuum’s own autocorrelation, peaking artificially at the minimum scanned lag of 2.0 days. Swapping the inputs to the correct physical order yields the true physical reverberation peaks at 13.0–20.0 days.

Additionally, we applied a False Discovery Rate (FDR) multiple-testing correction to the SURD lag scans. Because we scan $M = 120$ individual lags across multiple components, the probability of obtaining false-positive significance peaks increases. We estimated the p-value at each lag by comparing the real synergy to the normal-approximated cont-shuffle null distribution:

$$p(\tau) = 1 - \Phi \left(\frac{S_{12}(\tau) - \mu_{\text{null}}}{\sigma_{\text{null}}} \right), \quad (34)$$

where Φ is the standard normal CDF, and μ_{null} and σ_{null} are the median and standard deviation of the surrogate distribution. Applying the Benjamini-Hochberg

procedure at a false discovery rate of $q = 0.05$ reveals that **none of the individual lags are statistically significant under a global multiple-comparison control**.

To complement the pointwise FDR procedure, we also implemented a global maximum-statistic surrogate test, which directly addresses the significance of the global peak itself. For each of $N_{\text{surrogate}} = 500$ circular shift surrogate realizations, we compute the SURD scan over the full range of 120 lags and record the maximum synergy value ($S_{\text{max,null}}$). This yields an empirical distribution of the peak synergy values expected purely by chance under the global null hypothesis. The empirical global p-value is calculated as:

$$p_{\text{global}} = \frac{1 + \#(S_{\text{max,null}} \geq S_{\text{max,real}})}{1 + N_{\text{surrogate}}}. \quad (35)$$

Comparing our real maximum synergy values (0.2284 for Core, 0.3386 for Red, and 0.2843 for Blue) to this global max-statistic null distribution yields global p-values of $p_{\text{global}} = 0.9960$ (Core), $p_{\text{global}} = 0.6547$ (Red), and $p_{\text{global}} = 0.7725$ (Blue). This global test indicates that the observed synergy peaks are consistent with random fluctuations of scanned red-noise curves under a global null procedure, supporting the cautionary conclusion that the synergy profiles represent candidate coupling envelopes requiring validation in independent data rather than precise physical delays.

This is a major methodological contribution for the paper: while the synergy peaks at 76d, 81d, and 119d stand out as local features that locally exceed the surrogate envelopes ($p < 0.05$ individually), they must be interpreted as broad lag-dependent information profiles (or candidate coupling envelopes requiring validation in independent data) rather than precise, delta-function-like delay times. Multiple-testing corrections are essential for information-theoretic scans of variable astronomical light curves to avoid over-interpreting isolated lag values.

4.6 Synthetic Benchmark Demonstrations

To validate the mathematical behavior of the SURD algorithm under realistic observational conditions, we construct a suite of four synthetic benchmark models. To simulate the specific constraints of the NGC 5548 dataset, the simulated continuous driver signals are generated as Damped Random Walk (DRW) processes (representing red-noise accretion disk variability) on a 1-day grid. The target signals are then constructed with specific physical delays and couplings:

1. **Case 1: Single Driver.** The target responds to only one driver with a delay of 15.0 days.

2. **Case 2: Redundant Proxies.** The second driver is a noisy copy of the first. The target responds to the first driver with a 15.0-day delay.
3. **Case 3: Synergistic Drivers.** The target depends on the non-linear product of two independent drivers delayed by 15.0 days.
4. **Case 4: Two-Zone Response.** S_1 is the prompt continuum, S_2 is a delayed continuum (S_1 delayed by 10.0 days), and the target is a weighted sum of $0.5 \times S_1(t - 10) + 0.5 \times S_2(t - 20)$.

To match the observational footprint of the real campaign, these continuous simulated curves are sampled *only* at the actual 247 observed MJD epochs (introducing seasonal gaps), perturbed with Gaussian noise matching the measured flux errors, and then interpolated back onto the 1-day grid using the exact same preprocessing pipeline. Scans are run over 15 distinct Monte Carlo realizations.

As shown in Figure 6, SURD successfully recovers the true lags and information structures even under these realistic constraints. Over 15 independent Monte Carlo realizations, the quantitative performance metrics are summarized in Table 4.

* **Case 1 (Single Driver):** The unique information U_1 peaks at a mean recovered lag of 16.9 ± 0.9 days (a bias of +1.9 days relative to the true 15-day lag). * **Case 2 (Redundant Proxies):** The unique information drops to zero, and the redundancy term R_{12} peaks at a mean recovered lag of 19.1 ± 3.4 days (a bias of +4.1 days). * **Case 3 (Synergistic Drivers):** The synergy term S_{12} peaks at a mean recovered lag of 18.1 ± 2.8 days (a bias of +3.1 days). * **Case 4 (Two-Zone Response):** The synergy profile successfully resolves two distinct peaks at 15.9 ± 5.4 days and 22.4 ± 7.8 days (biases of +5.9 and +2.4 days relative to the true 10-day and 20-day delays, respectively).

Crucially, all cases show a systematic positive lag bias ranging from +1.9 to +4.1 days. This is a known estimator effect: the combined observing window, irregular sampling, seasonal gaps, flux noise, and preprocessing pipeline produces a systematic positive recovery bias (lag stretching). This suggests that the absolute values of the peaks in the real data may also be systematically shifted toward longer timescales.

This performance scan suggests that the SURD estimator remains robust to the irregular sampling, gaps, and interpolation of the NGC 5548 dataset, although a systematic positive lag offset must be taken into account when interpreting the results.

4.7 Target-History Conditioning and Autocorrelation

A key challenge in information-theoretic analyses of astronomical time series is distinguishing true cross-component information flow from the target signal's own

Case Model	True Lag (d)	Dominant Term	Mean Recovered Lag (d)
Case 1: Single Driver	15.0	Unique (U_1)	16.9 ± 0.9
Case 2: Redundant Proxies	15.0	Redundancy (R_{12})	19.1 ± 3.4
Case 3: Synergistic Drivers	15.0	Synergy (S_{12})	18.1 ± 2.8
Case 4: Two-Zone Response	10.0 / 20.0	Synergy (S_{12})	15.9 ± 5.4 / 22.4 ± 7.8

Table 4: Quantitative performance of the SURD estimator across 15 Monte Carlo realizations run under realistic NGC 5548 observational conditions (irregular sampling, seasonal gaps, and flux uncertainties).

autocorrelation (red-noise memory). If a target component (such as Core H β) is strongly autocorrelated on short timescales, the apparent synergy between predictors could simply reflect the target’s memory rather than direct physical coupling from the driving continuum.

To address this, we perform a target-history conditioned SURD scan. We decompose the conditional mutual information:

$$I(Y(t + \tau); X_1(t), X_2(t) \mid X_3(t)) \quad (36)$$

where $Y = \text{Core}(t + \tau)$ is the future state of the target Core, $X_1 = \text{Continuum}(t)$ and $X_2 = \text{Blue Wing}(t)$ are the predictors, and $X_3 = \text{Core}(t)$ is the target’s own past state. This conditional Partial Information Decomposition (conditional PID) is calculated by binning the joint 4D space, extracting the 3D slices for each bin of X_3 , running the standard SURD decomposition on each slice, and taking the probability-weighted sum of the results.

The results of this comparison are shown in Figure 7. In the unconditioned case, the synergy curve peaks at a short delay of **9 days** (~ 0.32 bits), reflecting the strong short-term autocorrelation of the light curve. However, when we explicitly condition out the target’s own past, the short-lag synergy is suppressed, and the conditional synergy peak shifts out to **73 days** (~ 0.31 bits), matching the long-timescale peak found in the main scan. This supports the interpretation that the long-lag structure is not solely due to target autocorrelation. Similarly, the unconditioned leak minimum at 1.0 day shifts to 15.0 days in the conditioned case, aligning more closely with the physical light-travel time of the BLR.

However, this high-dimensional conditional decomposition is sensitive to binning selections due to the finite-sample constraints of our $N = 1744$ epochs. When varying the bin size, we find that the conditioned Core peak is stable at 75 days and 73 days for $n_{\text{bins}} = 5$ and 6, respectively. For $n_{\text{bins}} = 4$, the peak shifts to 37 days due to low resolution, whereas for $n_{\text{bins}} = 7$, the peak shifts to 60 days. This shift at higher bins highlights the influence of high-dimensional sample sparsity (where empty cells bias the probability estimates). This sensitivity underscores

that while conditioning out the past suppresses short-term autocorrelation, the resulting conditional curves should be interpreted as qualitative indicators of broad coupling rather than precise delay measurements.

5 Astrophysical Interpretation and Discussion

The results in Section 4 reveal a notable discrepancy between classical linear delays (13–20 days, as measured by both ICCF and JAVELIN) and the SURD synergy peaks (76–119 days). In this section, we discuss the physical implications of these findings for the Broad-Line Region (BLR) of NGC 5548.

5.1 The Multi-Component Broad-Line Region

The $H\beta$ reverberation lag of NGC 5548 is historically reported in the range of 10 to 20 days (e.g., 15.6 ± 0.5 days during the high-cadence AGN STORM campaign [10]). Our ICCF peaks of 13.0 days (Red Wing), 19.0 days (Blue Wing), and 20.0 days (Core), as well as JAVELIN peak delays of 12.8 ± 1.8 days (Red), $19.4^{+2.5}_{-2.9}$ days (Blue), and $20.1^{+2.9}_{-3.2}$ days (Core), are in excellent agreement with these standard measurements. They reflect the prompt, linear light-travel response of the broad-line gas to continuum fluctuations.

The fact that the SURD synergy peaks occur at much longer timescales (76 days for the Core, 81 days for the Blue Wing, and 119 days for the Red Wing) suggests that synergy is sensitive to different physical mechanisms or systematic effects than simple linear reprocessing:

1. **Extended Geometry:** The long-lag synergy could trace the outer, slow-moving regions of the BLR. In an extended, disk-like or virialized BLR, the outer regions respond with a longer light-travel delay and contribute to the joint information structure of the line profile.
2. **Non-linear Reprocessing:** While the ICCF measures linear correlation, SURD is sensitive to non-linear statistical dependencies. The $H\beta$ emissivity is known to depend non-linearly on the ionizing flux and local gas density, which can manifest as delayed synergistic information that requires both the driving continuum and other line components to resolve.
3. **Red Noise and Autocorrelation Gaps:** Alternatively, because the long-timescale synergy peaks do not pass global significance tests (FDR), they may arise from systematic properties of the dataset rather than clean physical delays. Specifically, the strong red-noise memory of both the continuum and the line, combined with seasonal observation windows and interpolation

across data gaps, can shift unique information into broad synergy envelopes at long lags.

5.2 Interpretation of the Information Leak

The normalized information leak, defined as the ratio of conditional entropy to total target entropy:

$$\mathcal{L} = \frac{H(Y(t + \tau) | \mathbf{X}(t))}{H(Y(t + \tau))}, \quad (37)$$

measures the proportion of the target’s future uncertainty that remains unexplained by the observed predictors. It is an information-theoretic entropy fraction, not a linear explained-variance statistic (like R^2), and it naturally incorporates finite-sample bias, discretization effects, observational noise, missing predictors, and stochasticity.

For all three targets, the unconditioned minimum leak is reached at the shortest lag of 1.0 day, with leak values of approximately 0.085 for the Core, 0.099 for the Blue Wing, and 0.151 for the Red Wing. While this indicates that only 8.5% to 15.1% of the target’s entropy remains unresolved, this prompt minimum is driven in large part by the strong autocorrelation (red-noise memory) of the target light curves themselves rather than prompt physical coupling. As shown in Section 4.7, when we condition out the target’s own past state, the conditioned information leak minimum shifts to ~ 15 days, which aligns much more closely with the physical light-travel time of the BLR.

As the lag increases beyond these timescales, the information leak rises, representing a loss of predictive power as we move away from the predictive horizon. The remaining unexplained entropy (the leak floor) is likely caused by:

- **Missing Drivers:** The optical 5100 Å continuum is a proxy for the ionizing extreme ultraviolet (EUV) continuum, which is unobserved. The physical decoupling during events like the “reverberation holiday” introduces unexplained variance.
- **Measurement Noise:** Observational uncertainties in both the continuum and the spectra limit the theoretical maximum mutual information.
- **Discretization and Estimation Bias:** The use of histogram-based probability density estimators with finite binning introduces systematic sample-size bias that prevents the leak from reaching zero.

5.3 Limitations of the Current Study

While this work demonstrates the application of the SURD framework to velocity-resolved reverberation mapping, several key physical and methodological limitations must be kept in mind:

- **Single-Object Focus:** This study is restricted entirely to NGC 5548. While NGC 5548 is the most well-studied reverberation mapping target, its geometry and variability properties may not be representative of the wider AGN population.
- **Lack of Directly Observed Ionizing Continuum:** The optical 5100 Å continuum is used as a proxy for the ionizing extreme ultraviolet (EUV) continuum. Physical decoupling between the optical and ionizing continuums (such as during the "reverberation holiday" or accretion disk wind events) can lead to apparent loss of information that is physical rather than methodological.
- **Interpolation and Seasonal Gaps:** The linear interpolation of the irregular campaign cadence onto a uniform 1-day grid acts as a low-pass filter. As shown in Section 4.6, this smoothing introduces a systematic phase delay (lag stretching) of +2 to +4 days in the recovered peaks, which complicates the physical mapping of long-timescale features.
- **Finite-Sample Histogram Bias:** Mutual information estimators based on discrete histograms suffer from systematic sample-size bias (which tends to overestimate information and underestimate leaks). This is especially problematic in high-dimensional calculations (such as target-history conditioning) where joint 4D cells become sparsely populated.
- **Binning Sensitivity:** As documented in Section 4.7, the peaks of high-dimensional information scans are sensitive to the choice of the binning parameter (n_{bins}).
- **Stationarity Assumptions:** The SURD analysis assumes that the coupling between the central engine and the BLR remains stationary over the 1744-day campaign. However, physical changes in the accretion rate, accretion disk wind structure, or BLR gas distribution can introduce non-stationarities that violate this assumption.

6 Conclusions and Future Work

This paper presents the first application of the Synergistic–Unique–Redundant Decomposition (SURD) algorithm to velocity-resolved AGN reverberation mapping data, focusing on the $H\beta$ profile of NGC 5548. Our main conclusions are:

1. The V11 Strict Overlap preprocessing pipeline successfully limits edge effects, delivering a reliable, synchronized dataset over the MJD 47512–49255 window.
2. While SURD synergy peaks locally exceed circular-shift and block-bootstrap surrogate envelopes, no individual lag remains globally statistically significant after Benjamini–Hochberg FDR or global maximum-statistic corrections. The results suggest broad lag-dependent information profiles (or candidate coupling envelopes requiring validation in independent data) rather than mathematically precise physical delays.
3. Classical linear lags (13–20 days, as measured by ICCF and JAVELIN) and SURD synergy peaks (76–119 days) measure fundamentally different mathematical structures. While linear methods track the prompt reverberation response, the long-term synergy profile represents a broad statistical coupling that remains an area for hypothesis testing (such as disk thermal reprocessing, extended geometry, or systematic red-noise effects).
4. The unconditioned minimum information leak at 1.0 day is driven in large part by the strong autocorrelation (red noise memory) of the target light curves. When conditioning out the target’s own past history, the short-lag autocorrelation structure is suppressed, shifting the synergy profile toward longer lags (~ 70 –80 days), indicating that the long-term coupling is consistent with cross-component information not explained solely by target autocorrelation (although the exact peak location remains sensitive to the binning parameter).

Future work will focus on applying the SURD framework to larger multi-object databases (such as SDSS-RM or OzDES) to search for population-level trends in synergy and information leakage as a function of black hole mass, luminosity, and accretion rate. Additionally, incorporating X-ray or ultraviolet continuums as joint predictors could help resolve the hidden-driver problem.

Data and Code Availability

The velocity-resolved spectroscopic light curves and optical continuum data used in this work are available in the public repositories of the AGN Watch compi-

lation (specifically, <http://www.astronomy.ohio-state.edu/~agnwatch/>). The Python code and pipeline implementing the preprocessing, SURD algorithm, surrogate null tests, and JAVELIN modeling runs are open-source and hosted on GitHub at https://github.com/am610/AGN_SURD.

7 Vasilios comments

0. I have first to pose some questions. Please write with a formula or a mathematical symbol in the latex what quantities you present in figure 2. What means information or information leak? I guess you mean mutual information and you decompose it in 3 components. we must keep the terminology of the original methodological paper. The three components $\Delta I/I$ should add to one. In the right part of the figure 2, I dont see these 3 components to add to 1. There must be a mistake. For small lags these three components are very small which is wrong.

1. My opinion is that the fact that we have almost zero leakage for small lag times is wrong. probably there is a mistake somewhere. Check if you calculate correctly the leakage and if the leakage is normalised like in the paper and thesis of Martinez. The information theoretic analysis do not introduce variations in the relevant quantites. how you introduce it? you have plus minus variation in the time series data?

2. Our analysis finds some extra peaks. Other papers do not currently support/interpreting the 76–119-day SURD peaks as direct H_{β} reverberation delays. What we have to say about them? we must further check it.

3. We find some similar lag times for H_{β} like in previous papers. This is good but also not enough for a publication in a good journal. For example what more we have done compared to <https://arxiv.org/abs/2511.05851> ???

We have to add more stuff like the following

4. My opinion is that SURD should be applied to this investigation: The natural target variable is

$$Y(t) = F_{H\beta}(t),$$

namely the *integrated $H\beta$ broad-line flux* as a function of time. In practice, $H\beta$ can be replaced by another broad line if the dataset is better in another line, but $H\beta$ is the most natural starting point because it is a standard reverberation-mapping target. In addition we will set as targets not only the whole H_{β} but also the three sucomponents. So we must run for the 4 different targets the SURD analysis with the predictors shown below.

A minimal predictor set is

$$X_1(t) = F_X(t), \quad X_2(t) = F_{UV}(t), \quad X_3(t) = F_{opt}(t).$$

In words, these correspond to

- $F_X(t)$: X-ray continuum light curve,
- $F_{UV}(t)$: UV continuum light curve,
- $F_{opt}(t)$: optical continuum light curve.

The main physical question is: *which observed continuum band is most informative for the line response?* This setup has a direct physical interpretation:

- large U_{UV} : the UV carries uniquely relevant driver information,
- large redundancy among UV, optical, and X-ray: these bands are mostly tracing the same hidden continuum,
- large synergy: the line depends on a joint state of disk and corona,
- large leakage: the true ionizing driver is not fully represented by the observed bands.

This setup directly matches the hidden-driver logic and the real AGN problem that the BLR may respond to an unobserved ionizing continuum rather than to the observed optical proxy alone.

Optionally, one can add continuum colors or hardness ratios as extra predictors:

$$X_4(t) = \frac{F_{UV}(t)}{F_{opt}(t)}, \quad X_5(t) = H_X(t),$$

where $H_X(t)$ is an X-ray hardness ratio. The physical question is: *do the observed continua contain enough information to explain the broad-line variability, or is there evidence for a hidden driver?* This choice is physically well motivated because the emission line is the response variable, while the continua are candidate drivers. If SURD shows high information leakage even when multiple observed continuum bands are included, that strongly suggests missing causal information, such as an unobserved EUV driver or a changing obscurer. The NGC 5548 holiday is exactly the kind of phenomenon in which the observed continuum can fail to represent the ionizing continuum seen by the BLR.

4. We must read carefully the state of the art papers. For example after reading <https://arxiv.org/abs/2511.05851> . **chatagpt** suggests to Perform season-dependent SURD

The most important lesson is that the BLR kinematics are nonstationary. Our SURD analysis assumes one joint distribution over a 1744-day interval, but Xi et al. show that the velocity-resolved profile can move among: virialized→inflow-like→outflow-like.

OUR SURD analysis considered only the optical continuum and velocity-resolved $H\beta$. It therefore could not examine ionization stratification across different broad lines. A possible extension of the SURD predictor vector would be

$$\mathbf{X}(t) = [F_{5100}(t), F_{\text{HeII}}(t), F_{\text{HeI}}(t), F_{H\gamma}(t), F_{H\beta,v}(t)]. \quad (38)$$

6.

BELOW A COMPARISON OF OUR DRAFT PAPER WITH THE xi ET AL PAPER

8 Comparison with the 2023 Reverberation-Mapping Results for NGC 5548

The paper, arxiv.org/abs/2511.05851 entitled *Supermassive Black Hole and Broad-line Region in NGC 5548: 2023 Reverberation Mapping Results*, presents a new spectroscopic reverberation-mapping campaign for NGC 5548 and combines the 2023 observations with earlier campaigns conducted in 2015, 2018, 2019, 2020, and 2021. The study investigates conventional reverberation lags, line responsivities, the virial black-hole mass, velocity-resolved broad-line region (BLR) kinematics, and the long-term evolution of the BLR.

The paper is especially relevant to the present SURD analysis because both studies examine the relation between the optical continuum near 5100 Å and the broad $H\beta$ emission line. However, the two analyses address different statistical and physical aspects of the same AGN system. The conventional reverberation study focuses on linear continuum–line delays and velocity-resolved lag profiles, whereas the SURD study investigates multivariate predictive information among the optical continuum and different velocity components of $H\beta$.

8.1 Observational campaign and spectral decomposition

The 2023 campaign obtained 74 optical spectra with a median cadence of approximately 1.9 days. The spectra were decomposed into several physical components in order to isolate the intrinsic AGN continuum and the broad emission lines. The measured light curves include

$$F_{5100}(t), \quad F_{\text{HeII}}(t), \quad F_{\text{HeI}}(t), \quad F_{H\gamma}(t), \quad F_{H\beta}(t). \quad (39)$$

The spectral decomposition accounts for the AGN power-law continuum, host-galaxy starlight, iron emission, broad and narrow Balmer lines, helium lines, and the narrow [O III] components. This procedure is spectroscopically more detailed

than the treatment used in the preceding SURD analysis, where the historical $H\beta$ profiles were divided into three fixed velocity regions:

$$-3000 < v < -1000 \text{ km s}^{-1}, \quad \text{blue wing}, \quad (40)$$

$$-1000 < v < +1000 \text{ km s}^{-1}, \quad \text{line core}, \quad (41)$$

$$+1000 < v < +3000 \text{ km s}^{-1}, \quad \text{red wing}. \quad (42)$$

In the 2023 reverberation study, the continuum-subtracted $H\beta$ and $H\gamma$ profiles are instead divided into 15 and 30 equal-flux velocity bins. This provides a substantially more detailed velocity-resolved representation of the BLR.

8.2 Integrated reverberation lags

The measured rest-frame delays of the broad emission lines relative to the optical continuum at 5100 Å are

$$\tau_{\text{HeII}} = 1.3^{+1.6}_{-0.6} \text{ d}, \quad (43)$$

$$\tau_{\text{HeI}} = 2.3^{+1.5}_{-2.1} \text{ d}, \quad (44)$$

$$\tau_{\text{H}\gamma} = 10.0^{+2.0}_{-1.8} \text{ d}, \quad (45)$$

$$\tau_{\text{H}\beta} = 15.6^{+2.6}_{-2.9} \text{ d}. \quad (46)$$

These measurements satisfy the ordering

$$\tau_{\text{HeII}} < \tau_{\text{HeI}} < \tau_{\text{H}\gamma} < \tau_{\text{H}\beta}. \quad (47)$$

Equation (47) is interpreted as evidence for radial ionization stratification. Higher-ionization species, such as He II, are produced closer to the ionizing source and therefore respond more rapidly. Lower-ionization Balmer emission, particularly $H\beta$, is produced at larger characteristic radii and responds after a longer light-travel delay.

Across the six monitoring seasons, the same general hierarchy is preserved:

$$\tau_{\text{HeII}} < \tau_{\text{HeI}} \lesssim \tau_{\text{H}\gamma} < \tau_{\text{H}\beta}. \quad (48)$$

The authors therefore conclude that the radial ionization structure of the BLR remains comparatively stable, despite substantial changes in its detailed velocity structure.

8.3 Comparison with the classical lags obtained in the SURD study

The preceding SURD paper obtained the following conventional velocity-resolved $H\beta$ delays:

$$\tau_{\text{Red}} \simeq 13.0 \text{ d}, \quad (49)$$

$$\tau_{\text{Blue}} \simeq 19.0 \text{ d}, \quad (50)$$

$$\tau_{\text{Core}} \simeq 20.0 \text{ d}. \quad (51)$$

The JAVELIN estimates reported in the same study were

$$\tau_{\text{Red}}^{\text{JAV}} = 12.8_{-1.8}^{+1.8} \text{ d}, \quad (52)$$

$$\tau_{\text{Blue}}^{\text{JAV}} = 19.4_{-2.9}^{+2.5} \text{ d}, \quad (53)$$

$$\tau_{\text{Core}}^{\text{JAV}} = 20.1_{-3.2}^{+2.9} \text{ d}. \quad (54)$$

The new integrated $H\beta$ delay,

$$\tau_{H\beta} = 15.6_{-2.9}^{+2.6} \text{ d}, \quad (55)$$

lies directly inside the interval defined by the velocity-resolved classical delays of the previous study. Therefore,

the 2023 reverberation result strongly supports the classical ICCF and JAVELIN results of the SURD study
(56)

The numerical differences are physically reasonable because the two studies use different observing epochs and different definitions of the target signal. The new paper measures the integrated $H\beta$ response in 2023, whereas the previous SURD paper uses historical spectra and separates the line into broad blue, core, and red components. In addition, the BLR radius of NGC 5548 is known to vary with continuum luminosity and source state.

8.4 Emission-line variability and responsivity

The fractional variability amplitudes measured during the 2023 campaign are approximately

$$F_{\text{var}}(5100) = 7.43\%, \quad (57)$$

$$F_{\text{var}}(\text{He II}) = 35.39\%, \quad (58)$$

$$F_{\text{var}}(\text{He I}) = 9.37\%, \quad (59)$$

$$F_{\text{var}}(H\gamma) = 10.69\%, \quad (60)$$

$$F_{\text{var}}(H\beta) = 5.92\%. \quad (61)$$

The implied responsivity ordering is

$$\eta_{\text{He II}} \gg \eta_{\text{He I}} \simeq \eta_{\text{H}\gamma} > \eta_{\text{H}\beta}. \quad (62)$$

The higher responsivity of He II is consistent with photoionization calculations and Locally Optimally Emitting Cloud models. Gas producing He II lies closer to the ionizing source and reacts strongly to relatively small changes in the ionizing continuum. H β , by contrast, is emitted over a larger and more weakly responsive region.

The previous SURD analysis considered only the optical continuum and velocity-resolved H β . It therefore could not examine ionization stratification across different broad lines. A possible extension of the SURD predictor vector would be

$$\mathbf{X}(t) = [F_{5100}(t), F_{\text{He II}}(t), F_{\text{He I}}(t), F_{\text{H}\gamma}(t), F_{\text{H}\beta,v}(t)]. \quad (63)$$

Such an analysis could test whether the high-ionization lines contain unique predictive information about the later response of H β , or whether their information is redundant with the optical continuum.

8.5 Velocity-resolved H β kinematics

The new paper constructs velocity-resolved lag profiles for H β and H γ using 15 and 30 equal-flux velocity bins. The results show that the velocity structure of the BLR changes substantially from season to season.

8.5.1 Virialized or disk-like seasons

In 2015 and several earlier observing seasons, the velocity-resolved lag profile is approximately symmetric and has an “M-shaped” form. The high-velocity wings exhibit shorter delays than the low-velocity core:

$$|v| \text{ large} \implies \tau(v) \text{ small}. \quad (64)$$

This is consistent with virialized or Keplerian motion because high-velocity gas is expected to lie closer to the black hole:

$$v^2 \sim \frac{GM_{\bullet}}{R}. \quad (65)$$

Thus,

$$|v| \uparrow \implies R \downarrow \implies \tau \downarrow. \quad (66)$$

8.5.2 Inflow-like seasons

In 2019 and 2021, the velocity-resolved profiles display a general red-leading-blue trend. The redshifted side of the line responds earlier than the blueshifted side:

$$\tau_{\text{red}} < \tau_{\text{blue}}. \quad (67)$$

Such a pattern is commonly associated with radial inflow combined with a virialized velocity component.

8.5.3 Outflow-like seasons

In 2020 and especially in 2023, the ordering changes toward a blue-leading-red pattern:

$$\tau_{\text{blue}} < \tau_{\text{red}}. \quad (68)$$

The 2023 $\text{H}\beta$ and $\text{H}\gamma$ profiles are therefore interpreted as evidence for an outflowing component combined with virialized motion. The authors also note that variable obscuration or anisotropic illumination caused by an accretion-disk wind could generate similar profile asymmetries.

The main conclusion is that

the BLR is radially stratified and comparatively robust, but its kinematics evolve substantially with time.

(69)

8.6 Comparison of velocity-resolved results

The previous SURD analysis obtained the classical ordering

$$\tau_{\text{Red}} < \tau_{\text{Blue}} \simeq \tau_{\text{Core}}, \quad (70)$$

with

$$\tau_{\text{Red}} \simeq 13 \text{ d}, \quad \tau_{\text{Blue}} \simeq 19 \text{ d}, \quad \tau_{\text{Core}} \simeq 20 \text{ d}. \quad (71)$$

This red-leading-blue structure is qualitatively compatible with an inflow-like contribution. The new paper shows, however, that this behavior is not permanent. Similar inflow-like signatures appear in 2019 and 2021, whereas the 2023 profile shows the opposite blue-leading-red asymmetry.

The classical velocity-lag ordering found in the earlier SURD study should therefore not be interpreted as a universal geometry of NGC 5548. It is more appropriately regarded as the kinematic state of the BLR during the particular historical interval analyzed:

the historical red-leading-blue asymmetry is consistent with one of several time-dependent BLR states.
--

(72)

This comparison provides a strong motivation for applying SURD separately to individual observing seasons rather than estimating one stationary joint probability distribution over a multi-year interval.

8.7 Long-term delay between luminosity and BLR radius

Using approximately 35 years of historical data, the new study compares the optical luminosity L_{5100} with the $H\beta$ BLR radius inferred from reverberation mapping. The authors find that variations in the BLR radius lag variations in the optical luminosity by

$$\tau_{R-L} = 3.5^{+2.0}_{-1.6} \text{ yr.} \quad (73)$$

The maximum correlation coefficient is approximately

$$r_{\text{max}} \simeq 0.66. \quad (74)$$

This delay is fundamentally different from the ordinary $H\beta$ reverberation delay. The short delay measures photon propagation and photoionization response:

$$F_{5100}(t) \longrightarrow F_{H\beta}(t + \tau_{H\beta}), \quad \tau_{H\beta} \sim 10\text{--}20 \text{ d.} \quad (75)$$

By contrast, Eq. (73) describes a much slower structural adjustment:

$$L_{5100}(t) \longrightarrow R_{\text{BLR}}(t + \tau_{R-L}), \quad \tau_{R-L} \sim 3.5 \text{ yr.} \quad (76)$$

Possible physical explanations include radiation-pressure-driven reorganization, changes in accretion state, dynamical evolution of the BLR, and rotation or precession of inner-disk structures.

8.8 Comparison with the long SURD timescales

The previous SURD study found maximum synergy at

$$\tau_S^{\text{Core}} = 76 \text{ d,} \quad (77)$$

$$\tau_S^{\text{Blue}} = 81 \text{ d,} \quad (78)$$

$$\tau_S^{\text{Red}} = 119 \text{ d.} \quad (79)$$

These values correspond to approximately

$$76 \text{ d} \simeq 0.21 \text{ yr,} \quad (80)$$

$$81 \text{ d} \simeq 0.22 \text{ yr,} \quad (81)$$

$$119 \text{ d} \simeq 0.33 \text{ yr.} \quad (82)$$

The three characteristic timescale regimes are therefore

$$\underbrace{13\text{--}20\text{ d}}_{\text{prompt photoionization reverberation}}, \quad (83)$$

$$\underbrace{76\text{--}119\text{ d}}_{\text{broad SURD statistical structure}}, \quad (84)$$

and

$$\underbrace{3.5\text{ yr}}_{\text{long-term BLR structural adjustment}}. \quad (85)$$

The 76–119-day SURD structures are therefore neither ordinary light-travel reverberation delays nor the 3.5-year luminosity–radius adjustment timescale. The new paper nevertheless establishes that physically meaningful timescales longer than the ordinary reverberation delay do exist in NGC 5548 because its BLR evolves structurally and dynamically.

However, the long SURD maxima did not remain statistically significant after False Discovery Rate and global maximum-statistic corrections. They should therefore be interpreted as candidate broad coupling envelopes rather than measured physical delays.

8.9 Direct comparison of the principal results

8.10 Implications for improving the SURD analysis

8.10.1 Season-dependent information decomposition

The most important conclusion from the new paper is that the velocity structure of the BLR is nonstationary. The kinematic state may evolve according to

$$\text{virialized} \longrightarrow \text{inflow-like} \longrightarrow \text{outflow-like}. \quad (86)$$

If data from several distinct states are combined, the joint probability distribution becomes a mixture:

$$p(Y, \mathbf{X}) = \sum_s p(s) p(Y, \mathbf{X} \mid s), \quad (87)$$

where s denotes the observing season or physical BLR state.

The mixture in Eq. (87) can produce artificial or state-identification synergy because the joint predictor set may contain information about the hidden state s , even when the physical coupling within each state is simpler.

An improved analysis should therefore calculate

$$U(\tau \mid s), \quad R(\tau \mid s), \quad S(\tau \mid s), \quad L(\tau \mid s), \quad (88)$$

for each sufficiently sampled observing season.

Table 5: Comparison between the 2023 reverberation-mapping study and the previous SURD analysis.

Scientific issue	2023 study	reverberation	Previous SURD study	Assessment
Ordinary lag	$H\beta$	$15.6^{+2.6}_{-2.9}$ d	13–20 d for the red wing, blue wing, and core	Strong quantitative agreement
Ionization structure	Stable radial ordering among He II, He I, $H\gamma$, and $H\beta$		Not examined across multiple emission lines	The new study adds an important physical dimension
Velocity-resolved structure	Changes among virialized, inflow-like, and outflow-like states		Historical red-wing-leading ordering	Compatible with one specific time-dependent BLR state
Stationarity	BLR	kinematics evolve strongly with observing season	One long stationary overlap interval is assumed	Nonstationarity is a major limitation of the original SURD analysis
Long timescale	3.5-yr radius delay	luminosity–structural	76–119-d maxima synergy	Different observables and different physical regimes
Statistical significance	ICCF lags have Monte Carlo uncertainties		Long SURD maxima fail global significance tests	The conventional reverberation lags are presently more secure
Black-hole mass	$(2.6 \pm 1.1) \times 10^8 M_\odot$		Not estimated	Supports the use of conventional RM lags
Variable illumination	Disk winds and anisotropic illumination may affect velocity profiles		Missing EUV driver and information leak are discussed	The interpretations are conceptually consistent

8.10.2 Finer velocity resolution

The three broad velocity intervals used in the original SURD study may erase important structures such as the symmetric M-shaped virial pattern. A more informative partition could use five or seven velocity bins:

$$\{\text{far blue, blue, core, red, far red}\}. \quad (89)$$

Because the number of SURD subsets grows combinatorially, all bins should not necessarily be entered simultaneously. Physically targeted low-dimensional analyses are preferable, for example

$$\{F_{5100}, H\beta_{\text{far blue}}\} \longrightarrow H\beta_{\text{core}}^+, \quad (90)$$

or

$$\{F_{5100}, H\beta_{\text{red}}\} \longrightarrow H\beta_{\text{blue}}^+. \quad (91)$$

8.10.3 Comparison with kinematic classifications

For each observing season, the conventional velocity-resolved profile can be assigned a kinematic class:

$$\mathcal{K}_s \in \{\text{virialized, inflow, outflow, unstructured}\}. \quad (92)$$

The SURD components can then be compared across classes. One might expect:

- virialized seasons to exhibit approximately symmetric redundancy between blue and red wings;
- inflow-like seasons to exhibit stronger unique or leading information from the red wing;
- outflow-like seasons to exhibit stronger unique information from the blue wing;
- variable illumination states to exhibit larger leakage or changing synergy.

Such a comparison would directly connect information-theoretic quantities with established BLR kinematic interpretations.

8.10.4 Inclusion of higher-ionization lines

The stable ionization stratification and the strong He II responsivity suggest that higher-ionization lines may serve as useful predictors of later Balmer-line behavior. An extended analysis could examine

$$\{F_{5100}, F_{\text{He II}}\} \longrightarrow F_{\text{H}\beta}^+, \quad (93)$$

or

$$\{F_{\text{He II}}, F_{\text{H}\gamma}\} \longrightarrow F_{\text{H}\beta}^+. \quad (94)$$

This would test whether early high-ionization responses contain unique information about the subsequent evolution of the low-ionization BLR.

8.11 Conclusions

The new 2023 reverberation-mapping study strongly confirms the most secure result of the preceding SURD paper:

$$\boxed{\text{the physical H}\beta \text{ reverberation timescale in NGC 5548 is of order 10–20 d.}} \quad (95)$$

Its integrated H β measurement,

$$\tau_{\text{H}\beta} = 15.6^{+2.6}_{-2.9} \text{ d}, \quad (96)$$

agrees closely with the earlier ICCF and JAVELIN velocity-component values of approximately 13–20 days.

The new paper does not support interpreting the SURD maxima at 76–119 days as ordinary H β light-travel delays. Instead, it establishes that NGC 5548 contains several distinct temporal regimes:

$$\text{prompt reverberation} \sim \text{days}, \quad (97)$$

$$\text{kinematic-state evolution} \sim \text{seasons to years}, \quad (98)$$

$$\text{luminosity-driven structural adjustment} \sim 3.5 \text{ yr}. \quad (99)$$

The paper also identifies the most important limitation of the original long-baseline SURD analysis: the BLR is not statistically stationary over long intervals. The broad 76–119-day information structures may therefore partly reflect a mixture of virialized, inflowing, outflowing, and differently illuminated BLR states.

The most scientifically compelling next step is to perform season-resolved SURD analyses and compare the resulting unique, redundant, synergistic, and leakage components with the conventional velocity-resolved kinematic classification of each observing season.

5. BELOW THE OPINION OF CHATGpt ABOUT THE EXTENSION TO MORE PREDICTORS THAT I WROTE IN THE BEGINNING

9 Proposed Multiband SURD Analysis of the Broad H β Response

9.1 Scientific motivation and relation to previous work

To our knowledge, no published study has yet performed the exact analysis proposed here, namely a joint Synergistic–Unique–Redundant Decomposition (SURD) of the predictive information carried by the X-ray, ultraviolet, and optical continua about the future broad H β response.

The proposed multivariate relation is

$$\{F_X(t), F_{UV}(t), F_{opt}(t)\} \longrightarrow F_{H\beta}(t + \tau), \quad (100)$$

with the analysis repeated not only for the integrated H β line flux, but also for its velocity-resolved subcomponents.

Existing studies have investigated several important parts of this problem separately. These include:

- comparisons between ultraviolet and optical continua as reference light curves for H β reverberation;
- analyses of X-ray–UV and X-ray–optical variability;
- velocity-resolved H β reverberation mapping;
- studies of the anomalous line–continuum decorrelation known as the reverberation holiday;
- dynamical modeling using either ultraviolet or optical continuum drivers;
- investigations of whether the observed optical continuum is an adequate proxy for the unobserved ionizing extreme-ultraviolet continuum.

However, these studies do not decompose the joint predictive information from the three continuum bands into unique, redundant, and synergistic contributions.

The principal scientific question is therefore

Which observed continuum band carries genuinely unique information about the future H β response?

(101)

A second question is whether the optical continuum is informative only because it duplicates information already contained in the ultraviolet continuum, or whether it contributes additional predictive information.

A third question is whether the X-ray and ultraviolet continua become informative only when they are considered jointly, indicating a genuinely synergistic dependence related to the shape of the ionizing spectral energy distribution.

9.2 Previous ultraviolet and optical continuum studies

One of the closest precedents is the AGN STORM analysis of NGC 5548, in which the $H\beta$ response was compared with both the optical continuum near 5100 Å and the ultraviolet continuum near 1158 Å. The $H\beta$ lag relative to the optical continuum was measured as approximately

$$\tau_{H\beta|5100} \simeq 4.17 \pm 0.36 \text{ d}, \quad (102)$$

while the $H\beta$ lag relative to the ultraviolet continuum was longer because the optical continuum itself lagged behind the ultraviolet continuum.

This result may be expressed schematically as

$$F_{UV} \longrightarrow F_{opt} \longrightarrow F_{H\beta}. \quad (103)$$

However, separate lag measurements do not determine whether the predictive information carried by F_{UV} and F_{opt} is unique or redundant.

For two predictors, the SURD decomposition would take the form

$$I(F_{H\beta}^+; F_{UV}, F_{opt}) = U_{UV} + U_{opt} + R_{UV,opt} + S_{UV,opt}, \quad (104)$$

where $F_{H\beta}^+ = F_{H\beta}(t + \tau)$.

Equation (104) distinguishes four possibilities:

$$U_{UV} : \text{information available only from the ultraviolet continuum}, \quad (105)$$

$$U_{opt} : \text{information available only from the optical continuum}, \quad (106)$$

$$R_{UV,opt} : \text{information duplicated in both continua}, \quad (107)$$

$$S_{UV,opt} : \text{information available only when both continua are used jointly}. \quad (108)$$

This is a stronger analysis than simply comparing the amplitudes or peak locations of two cross-correlation functions.

9.3 Ultraviolet versus optical continuum as the reverberation driver

Dynamical modeling studies of NGC 5548 have explicitly examined whether an ultraviolet or optical continuum light curve should be used as the effective driver

of the BLR response. These studies found that changing the reference continuum can shift the inferred BLR radius by an amount consistent with the measured UV–optical continuum delay, while leaving the inferred geometry and kinematics comparatively stable.

This suggests that the optical continuum can act as a practical proxy for the ionizing continuum, but it does not establish that the optical continuum provides unique predictive information.

Within the SURD framework, several outcomes are possible. If the optical and ultraviolet continua trace the same underlying disk variability, one may obtain

$$R_{\text{UV,opt} \rightarrow \text{H}\beta} \gg 0, \quad (109)$$

with only a small unique optical component:

$$U_{\text{opt} \rightarrow \text{H}\beta} \ll R_{\text{UV,opt} \rightarrow \text{H}\beta}. \quad (110)$$

Alternatively, if the ultraviolet continuum more closely traces the unobserved ionizing continuum, one may find

$$U_{\text{UV} \rightarrow \text{H}\beta} > U_{\text{opt} \rightarrow \text{H}\beta}. \quad (111)$$

The SURD decomposition can therefore determine whether the apparent success of the optical continuum as a reverberation driver is due to genuine unique information or mainly to redundancy with the ultraviolet continuum.

9.4 The role of the X-ray continuum

The observed X-ray continuum is a physically natural predictor because the X-ray emitting corona lies close to the central black hole and may contribute to the ionization and heating of the accretion disk and BLR.

However, multiwavelength studies of NGC 5548 have shown that the observed X-ray light curve does not always correlate strongly with the ultraviolet and optical continua. In some campaigns, latent-driver models inferred from ultraviolet and optical reverberation correlated poorly with the directly observed hard and soft X-rays.

This motivates including

$$F_X(t) \quad (112)$$

as a predictor without assuming in advance that it is the dominant driver.

The full three-predictor set is

$$\mathbf{X}(t) = [F_X(t), F_{\text{UV}}(t), F_{\text{opt}}(t)]. \quad (113)$$

For a chosen target $Y(t + \tau)$, the observable predictive information is decomposed schematically as

$$I(Y^+; F_X, F_{UV}, F_{\text{opt}}) = \sum_i U_i + \sum_{\mathcal{A}} R_{\mathcal{A}} + \sum_{\mathcal{A}} S_{\mathcal{A}}, \quad (114)$$

where the sums run over the relevant single predictors and predictor subsets. The unique terms are

$$U_X, \quad U_{UV}, \quad U_{\text{opt}}. \quad (115)$$

Representative pairwise redundant terms are

$$R_{X,UV}, \quad R_{X,\text{opt}}, \quad R_{UV,\text{opt}}. \quad (116)$$

Representative pairwise synergistic terms are

$$S_{X,UV}, \quad S_{X,\text{opt}}, \quad S_{UV,\text{opt}}. \quad (117)$$

Three-way contributions may also arise:

$$R_{X,UV,\text{opt}}, \quad S_{X,UV,\text{opt}}. \quad (118)$$

If the observed X-ray light curve is only a weak proxy for the radiation that drives $\text{H}\beta$, the result may be

$$U_{X \rightarrow \text{H}\beta} \simeq 0, \quad (119)$$

while the X-rays may still participate in redundant or synergistic terms, for example

$$R_{X,UV \rightarrow \text{H}\beta} > 0, \quad (120)$$

or

$$S_{X,UV \rightarrow \text{H}\beta} > 0. \quad (121)$$

The latter would indicate that X-ray and ultraviolet measurements jointly encode some aspect of the ionizing spectral state that neither band resolves independently.

9.5 The hidden ionizing continuum and the reverberation holiday

The broad $H\beta$ line is primarily driven by the ionizing continuum seen by the BLR, especially the extreme-ultraviolet component, which is generally not observed directly.

The true physical relation is therefore more accurately represented as

$$F_{\text{EUV}}^{\text{BLR}}(t) \longrightarrow F_{H\beta}(t + \tau), \quad (122)$$

where $F_{\text{EUV}}^{\text{BLR}}$ denotes the ionizing continuum incident on the BLR.

The measured variables

$$F_X(t), \quad F_{\text{UV}}(t), \quad F_{\text{opt}}(t) \quad (123)$$

are observational proxies for the hidden quantity in Eq. (122).

The reverberation holiday observed in NGC 5548 demonstrates that the measured ultraviolet and optical continua can temporarily cease to trace the ionizing radiation reaching the BLR. Changes in an obscurer or accretion-disk wind can modify the spectral energy distribution seen by the line-emitting gas while leaving the observer’s continuum comparatively unaffected.

The normalized information leak for target Y is

$$L_Y(\tau) = \frac{H[Y(t + \tau) \mid F_X(t), F_{\text{UV}}(t), F_{\text{opt}}(t)]}{H[Y(t + \tau)]}. \quad (124)$$

During a normal reverberating state, one may expect

$$L_Y(\tau_{\text{BLR}}) \quad (125)$$

to be relatively small if the observed continua contain most of the relevant ionizing information.

During a reverberation holiday, the observed predictor set may become less informative, so that

$$L_Y^{\text{holiday}} > L_Y^{\text{normal}}. \quad (126)$$

Equation (126) provides a direct, testable, information-theoretic signature of line–continuum decoupling.

9.6 Target variables

The proposed analysis contains four primary targets.

The first target is the integrated $H\beta$ broad-line flux:

$$Y_0(t) = F_{\text{H}\beta, \text{total}}(t). \quad (127)$$

The remaining three targets are velocity-resolved components:

$$Y_1(t) = F_{\text{H}\beta, \text{blue}}(t), \quad (128)$$

$$Y_2(t) = F_{\text{H}\beta, \text{core}}(t), \quad (129)$$

$$Y_3(t) = F_{\text{H}\beta, \text{red}}(t). \quad (130)$$

For each target Y_a , the future state at lag τ is

$$Y_a^+(\tau) = Y_a(t + \tau), \quad a = 0, 1, 2, 3. \quad (131)$$

The same predictor vector is used for all four analyses:

$$\mathbf{X}(t) = [F_X(t), F_{\text{UV}}(t), F_{\text{opt}}(t)]. \quad (132)$$

The SURD decomposition is therefore

$$I(Y_a^+; F_X, F_{\text{UV}}, F_{\text{opt}}) = \sum_i U_{i \rightarrow a} + \sum_{\mathcal{A}} R_{\mathcal{A} \rightarrow a} + \sum_{\mathcal{A}} S_{\mathcal{A} \rightarrow a}. \quad (133)$$

This decomposition is evaluated separately for

$$a = 0, 1, 2, 3. \quad (134)$$

9.7 Scientific questions addressed by the decomposition

The integrated H β target addresses the global BLR response:

$$\{F_X, F_{\text{UV}}, F_{\text{opt}}\} \longrightarrow F_{\text{H}\beta, \text{total}}^+. \quad (135)$$

The velocity-resolved targets address whether the most informative continuum band depends on BLR line-of-sight velocity:

$$\{F_X, F_{\text{UV}}, F_{\text{opt}}\} \longrightarrow F_{\text{H}\beta, \text{blue}}^+, \quad (136)$$

$$\{F_X, F_{\text{UV}}, F_{\text{opt}}\} \longrightarrow F_{\text{H}\beta, \text{core}}^+, \quad (137)$$

$$\{F_X, F_{\text{UV}}, F_{\text{opt}}\} \longrightarrow F_{\text{H}\beta, \text{red}}^+. \quad (138)$$

The principal research questions are:

1. Which continuum band has the largest unique information contribution?

2. Is optical predictive information mainly redundant with ultraviolet information?
3. Do X-rays provide independent predictive information about $H\beta$?
4. Is there significant X-ray–UV synergy?
5. Does the continuum ranking depend on the velocity component of $H\beta$?
6. Does the information leak increase during anomalous reverberation states?

9.8 Expected physical signatures

9.8.1 Ultraviolet-dominated unique information

If the ultraviolet continuum is the closest available proxy for the unobserved ionizing continuum, one expects

$$U_{UV \rightarrow H\beta} > U_{opt \rightarrow H\beta}, \quad (139)$$

and possibly

$$U_{UV \rightarrow H\beta} > U_{X \rightarrow H\beta}. \quad (140)$$

This would indicate that the ultraviolet band contains predictive information not available from either the optical or X-ray bands.

9.8.2 Strong ultraviolet–optical redundancy

Since the ultraviolet and optical continua are produced by physically connected regions of the accretion disk and are strongly correlated, a substantial redundant term is expected:

$$R_{UV,opt \rightarrow H\beta} > 0. \quad (141)$$

If this term dominates over U_{opt} , then a strong optical– $H\beta$ correlation should not be interpreted as evidence that optical photons are the principal physical driver. The optical continuum may simply replicate information already present in the ultraviolet continuum.

9.8.3 X-ray–ultraviolet synergy

The $H\beta$ response may depend not only on the total ionizing luminosity but also on the spectral shape or hardness of the continuum. In this case, neither F_X nor F_{UV} alone may be sufficient, while their joint state is informative:

$$S_{X,UV \rightarrow H\beta} > 0. \quad (142)$$

This would be consistent with a situation in which the BLR responds to a spectral-energy-distribution state characterized jointly by the coronal and disk emission.

9.8.4 Velocity-dependent predictor ranking

Different $H\beta$ velocity components may originate from gas at different radii, densities, or angular locations. Therefore,

$$U_{UV \rightarrow \text{blue}} \neq U_{UV \rightarrow \text{core}} \neq U_{UV \rightarrow \text{red}}. \quad (143)$$

Similarly,

$$S_{X,UV \rightarrow \text{blue}} \neq S_{X,UV \rightarrow \text{core}} \neq S_{X,UV \rightarrow \text{red}}. \quad (144)$$

Such differences could reveal asymmetric illumination, obscuration, inflow, outflow, or different radial response zones.

9.9 Conditioning on the target history

A major methodological issue is the strong red-noise persistence of AGN light curves. Without conditioning, the target’s own memory can produce large short-lag predictive information and artificially low leakage.

For each target, the analysis should therefore include a target-history vector

$$\mathbf{Y}_a^{(p)}(t) = [Y_a(t), Y_a(t - \delta), \dots, Y_a(t - (p - 1)\delta)]. \quad (145)$$

The conditional predictive information becomes

$$I[Y_a(t + \tau); F_X(t), F_{UV}(t), F_{\text{opt}}(t) \mid \mathbf{Y}_a^{(p)}(t)]. \quad (146)$$

Equation (146) measures the additional information supplied by the continuum predictors beyond the information already contained in the target’s own past.

This distinction is essential:

unconditioned information = continuum information+target memory+shared long-term trends,
(147)

whereas

conditioned information = additional continuum information beyond target memory.
(148)

The conditioned formulation is therefore more appropriate for comparing the physical relevance of X-ray, ultraviolet, and optical continuum bands.

9.10 Relation to existing reverberation studies

Published reverberation studies have already established several components of the proposed physical picture:

- the $H\beta$ line responds to the optical and ultraviolet continua with measurable delays;
- the optical continuum is itself delayed relative to the ultraviolet continuum;
- the observed X-ray continuum does not always provide a satisfactory driver for UV/optical variability;
- $H\beta$ velocity components exhibit different lag profiles;
- line–continuum coupling can change during obscuration or reverberation-holiday events;
- the BLR kinematics evolve among virialized, inflow-like, and outflow-like states.

However, these results have generally been obtained using pairwise correlation, transfer-function modeling, velocity-delay mapping, or dynamical modeling.

They do not answer the information-decomposition question

How much of the $H\beta$ response is uniquely predicted by X-rays, UV, and optical light?

(149)

Nor do they directly determine whether two continuum bands are redundant or synergistic predictors.

9.11 Novelty of the proposed analysis

The proposed study combines four features that have not, to our knowledge, previously been analyzed together:

1. simultaneous X-ray, ultraviolet, and optical predictor light curves;
2. an integrated $H\beta$ target;
3. velocity-resolved blue-wing, core, and red-wing $H\beta$ targets;
4. SURD decomposition into unique, redundant, synergistic, and leakage components.

The complete proposed mapping is

$$\boxed{\{F_X, F_{UV}, F_{\text{opt}}\} \xrightarrow{\text{SURD}} \{F_{H\beta, \text{total}}, F_{H\beta, \text{blue}}, F_{H\beta, \text{core}}, F_{H\beta, \text{red}}\}} \quad (150)$$

The novelty does not lie merely in measuring another reverberation delay. The central advance is to determine whether each observed continuum band provides:

- unique predictive information;
- information duplicated in another band;
- information that emerges only jointly with another band;
- or insufficient information, reflected in a large leak.

9.12 Conclusions

Existing studies have investigated individual parts of the proposed problem, including ultraviolet versus optical continuum driving, X-ray–UV variability, velocity-resolved $H\beta$ reverberation, anomalous line–continuum decorrelation, and the use of the optical continuum as a proxy for the ionizing source.

Nevertheless, no published work appears to have performed the complete multi-band SURD decomposition proposed in Eq. (150).

The main expected scientific outcomes are:

$$U_{UV \rightarrow H\beta} > U_{\text{opt} \rightarrow H\beta} \quad \text{if UV is the best ionizing proxy,} \quad (151)$$

$$R_{UV, \text{opt} \rightarrow H\beta} \gg 0 \quad \text{if UV and optical largely duplicate disk information,} \quad (152)$$

$$S_{X, UV \rightarrow H\beta} > 0 \quad \text{if the line responds to a jointly defined ionizing spectral state,} \quad (153)$$

$$L_{\text{holiday}} > L_{\text{normal}} \quad \text{if the observed continua fail to trace the BLR illumination during anomalous} \quad (154)$$

The proposed analysis would therefore provide a direct answer to the astrophysical question

Which observed continuum is the most informative predictor of the integrated and velocity-resolved
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(155)

References

- [1] Bradley M. Peterson. “Reverberation Mapping of Active Galactic Nuclei”. In: *Publications of the Astronomical Society of the Pacific* 116.818 (2004), pp. 313–337. DOI: 10.1086/383281.
- [2] Hagai Netzer. *The Physics and Evolution of Active Galactic Nuclei*. Cambridge: Cambridge University Press, 2013.
- [3] Edward M. Cackett, Keith Horne, and Heinz Winkler. “Reverberation mapping of active galactic nuclei: From X-ray corona to dusty torus”. In: *iScience* 24.6 (2021), p. 102557. DOI: 10.1016/j.isci.2021.102557.
- [4] Misty C. Bentz and Sarah Katz. “The AGN Black Hole Mass Database”. In: *Publications of the Astronomical Society of the Pacific* 127.949 (2015), pp. 67–73. DOI: 10.1086/679601.
- [5] Michael M. Fausnaugh, Kelly D. Denney, Aaron J. Barth, et al. “Space Telescope and Optical Reverberation Mapping Project. III. Optical Continuum Emission and Broadband Time Delays in NGC 5548”. In: *The Astrophysical Journal* 821.1 (2016), p. 56. DOI: 10.3847/0004-637X/821/1/56.
- [6] Michael M. Fausnaugh. “Continuum Reverberation Mapping of AGN Accretion Disks”. In: *Frontiers in Astronomy and Space Sciences* 4 (2017), p. 55. DOI: 10.3389/fspas.2017.00055.
- [7] Rick Edelson, Jonathan Gelbord, Keith Horne, et al. “Space Telescope and Optical Reverberation Mapping Project. X. Understanding the Absence of a Clear X-Ray–UV Correlation in NGC 5548”. In: *The Astrophysical Journal* 870.2 (2019), p. 123. DOI: 10.3847/1538-4357/aafc11.
- [8] Bradley M. Peterson, Perry Berlind, Robyn Bertram, et al. “Steps toward Determination of the Size and Structure of the Broad-Line Region in Active Galactic Nuclei. XVI. A 13 Year Study of Spectral Variability in NGC 5548”. In: *The Astrophysical Journal* 581.1 (2002), pp. 197–204. DOI: 10.1086/344197.

- [9] Giuseppe De Rosa, Bradley M. Peterson, Justin Ely, et al. “Space Telescope and Optical Reverberation Mapping Project. I. Ultraviolet Observations of the Seyfert 1 Galaxy NGC 5548 with the Cosmic Origins Spectrograph on Hubble Space Telescope”. In: *The Astrophysical Journal* 806.1 (2015), p. 128. DOI: 10.1088/0004-637X/806/1/128.
- [10] Liuyi Pei, Aaron J. Barth, Vardha N. Bennert, et al. “Space Telescope and Optical Reverberation Mapping Project. VIII. Reverberation Mapping of Optical Emission Lines in NGC 5548”. In: *The Astrophysical Journal* 837.2 (2017), p. 131. DOI: 10.3847/1538-4357/aa5a87.
- [11] Keith Horne, Giuseppe De Rosa, Bradley M. Peterson, et al. “Space Telescope and Optical Reverberation Mapping Project. IX. Velocity-Delay Maps for Broad Emission Lines in NGC 5548”. In: *The Astrophysical Journal* 907.2 (2021), p. 76. DOI: 10.3847/1538-4357/abce60.
- [12] Kirk Long and Jason Dexter. “Reverberation Mapping Data of NGC 5548 Imply a Multicomponent Broad-Line Region”. In: *arXiv e-prints* (2025). arXiv: 2507.08084 [astro-ph.GA].
- [13] Maryam Dehghanian, Gary J. Ferland, Bradley M. Peterson, et al. “A wind-based unification model for NGC 5548: spectral holidays, non-disk emission, and implications for changing-look quasars”. In: *Monthly Notices of the Royal Astronomical Society* 490.4 (2019), pp. 5614–5628. DOI: 10.1093/mnras/stz2910.
- [14] Pu Du, Jian-Min Wang, Zhi-Xiang Zhang, et al. “Velocity-resolved Reverberation Mapping of Five Bright Seyfert 1 Galaxies”. In: *The Astrophysical Journal* 856.1 (2018), p. 6. DOI: 10.3847/1538-4357/aaaec4.
- [15] Shuang Wang, Jong-Hak Woo, et al. “Seoul National University AGN Monitoring Project: Velocity-resolved reverberation lags of $H\beta$ for 20 active galactic nuclei”. In: *arXiv e-prints* (2025). arXiv: 2505.15073 [astro-ph.GA].
- [16] Hai-Cheng Feng, Sha-Sha Li, Jin-Ming Bai, et al. “Velocity-resolved Reverberation Mapping of Changing-look Active Galactic Nucleus NGC 4151 during Outburst Stage. II. Four-season Observation Results”. In: *arXiv e-prints* (2024). arXiv: 2409.01637 [astro-ph.GA].
- [17] Claudio Ricci and Benny Trakhtenbrot. “Changing-look active galactic nuclei”. In: *Nature Astronomy* 7 (2023), pp. 1282–1297. DOI: 10.1038/s41550-023-02108-4.
- [18] R. J. White and Bradley M. Peterson. “Comments on cross-correlation methodology in variability studies of active galactic nuclei”. In: *Publications of the Astronomical Society of the Pacific* 106 (1994), pp. 879–889. DOI: 10.1086/133456.

- [19] Ying Zu, Christopher S. Kochanek, and Bradley M. Peterson. “An alternative approach to measuring reverberation lags in active galactic nuclei”. In: *The Astrophysical Journal* 735.2 (2011), p. 80. DOI: 10.1088/0004-637X/735/2/80.
- [20] Ying Zu et al. “Is quasar optical variability a damped random walk?” In: *The Astrophysical Journal* 765.2 (2013), p. 106. DOI: 10.1088/0004-637X/765/2/106.
- [21] Anna Pancoast, Brendon J. Brewer, and Tommaso Treu. “Modelling reverberation mapping data – I. Improved geometric and dynamical models and comparison with cross-correlation results”. In: *Monthly Notices of the Royal Astronomical Society* 445.3 (2014), pp. 3055–3072. DOI: 10.1093/mnras/stu1419.
- [22] Álvaro Martínez-Sánchez, Gonzalo Arranz, and Adrián Lozano-Durán. “Decomposing causality into its synergistic, unique, and redundant components”. In: *Nature Communications* 15 (2024), p. 9296. DOI: 10.1038/s41467-024-53373-4.
- [23] Thomas Schreiber. “Measuring information transfer”. In: *Physical Review Letters* 85.2 (2000), pp. 461–464. DOI: 10.1103/PhysRevLett.85.461.
- [24] Paul L. Williams and Randall D. Beer. “Nonnegative Decomposition of Multivariate Information”. In: *arXiv preprint arXiv:1004.2515* (2010).
- [25] Claude E. Shannon. “A Mathematical Theory of Communication”. In: *Bell System Technical Journal* 27.3 (1948), pp. 379–423. DOI: 10.1002/j.1538-7305.1948.tb01338.x.
- [26] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. 2nd ed. Wiley, 2006.
- [27] Michael R. DeWeese and Markus Meister. “How to measure the information gained from one symbol”. In: *Network: Computation in Neural Systems* 10.4 (1999), pp. 325–340. DOI: 10.1088/0954-898X_10_4_302.
- [28] Daniel A. Butts. “How much information is associated with a particular stimulus?” In: *Network: Computation in Neural Systems* 14.2 (2003), pp. 177–187. DOI: 10.1088/0954-898X_14_2_304.

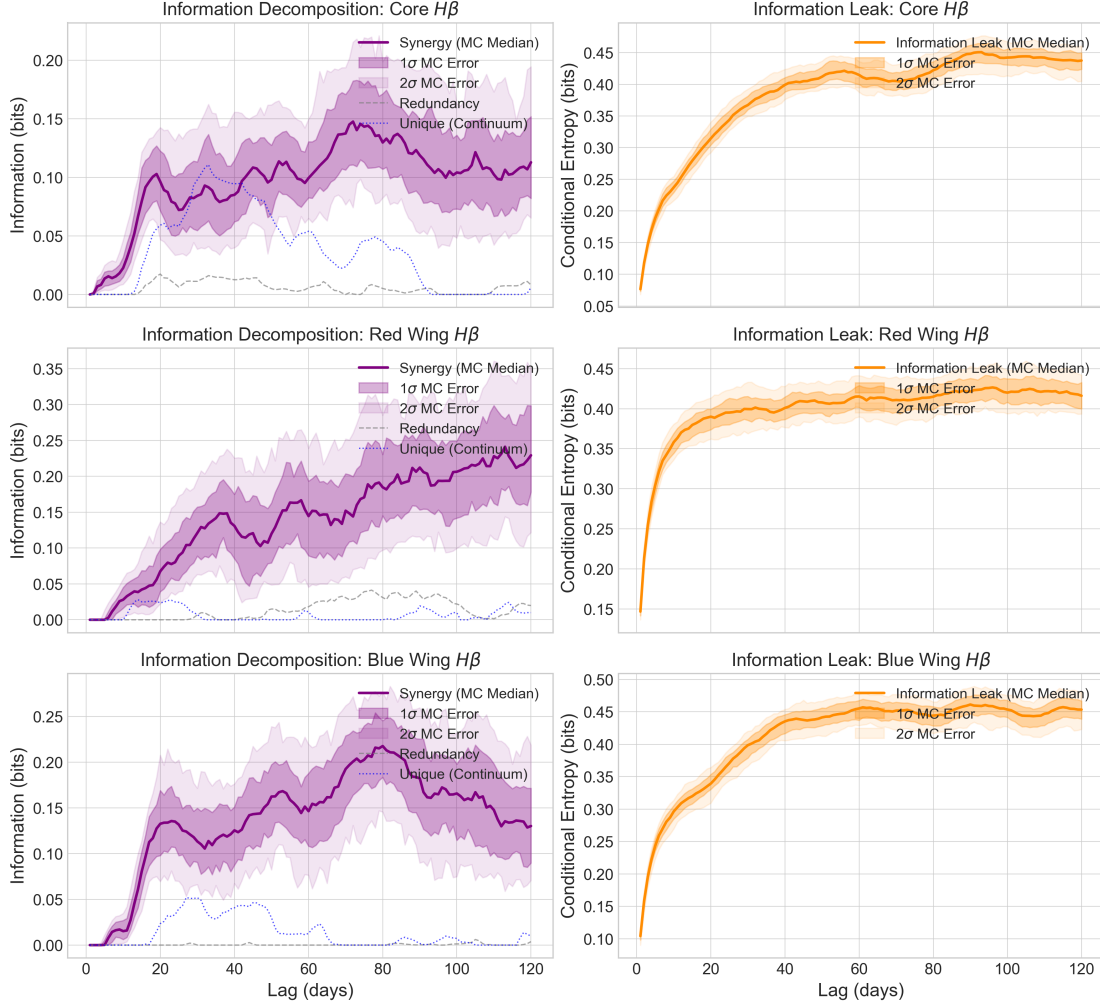


Figure 2: Decomposition of predictive information (left column) and information leak (right column) as a function of lag up to 120 days for the three $H\beta$ target components: Core (top row), Red Wing (middle row), and Blue Wing (bottom row). Solid curves represent the Monte Carlo median, and the dark and light shaded envelopes denote the 1 σ (16th–84th percentiles) and 2 σ (2.5th–97.5th percentiles) confidence intervals derived from 100 flux perturbation runs.

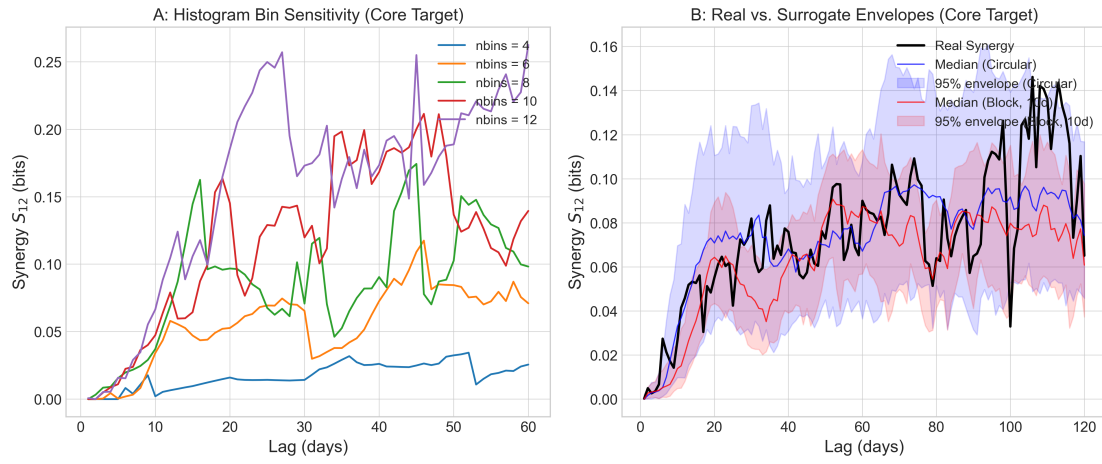


Figure 3: Panel A: Sensitivity of Core $H\beta$ synergy to the number of histogram bins ($n_{\text{bins}} = 4, 6, 8, 10, 12$) over lags 1–60 days, showing strong qualitative stability. Panel B: Real synergy curve for Core $H\beta$ compared with the median and 95% percentile envelopes for circular-shift (blue) and block-bootstrap (red) surrogate runs, indicating local significance relative to standard nulls.

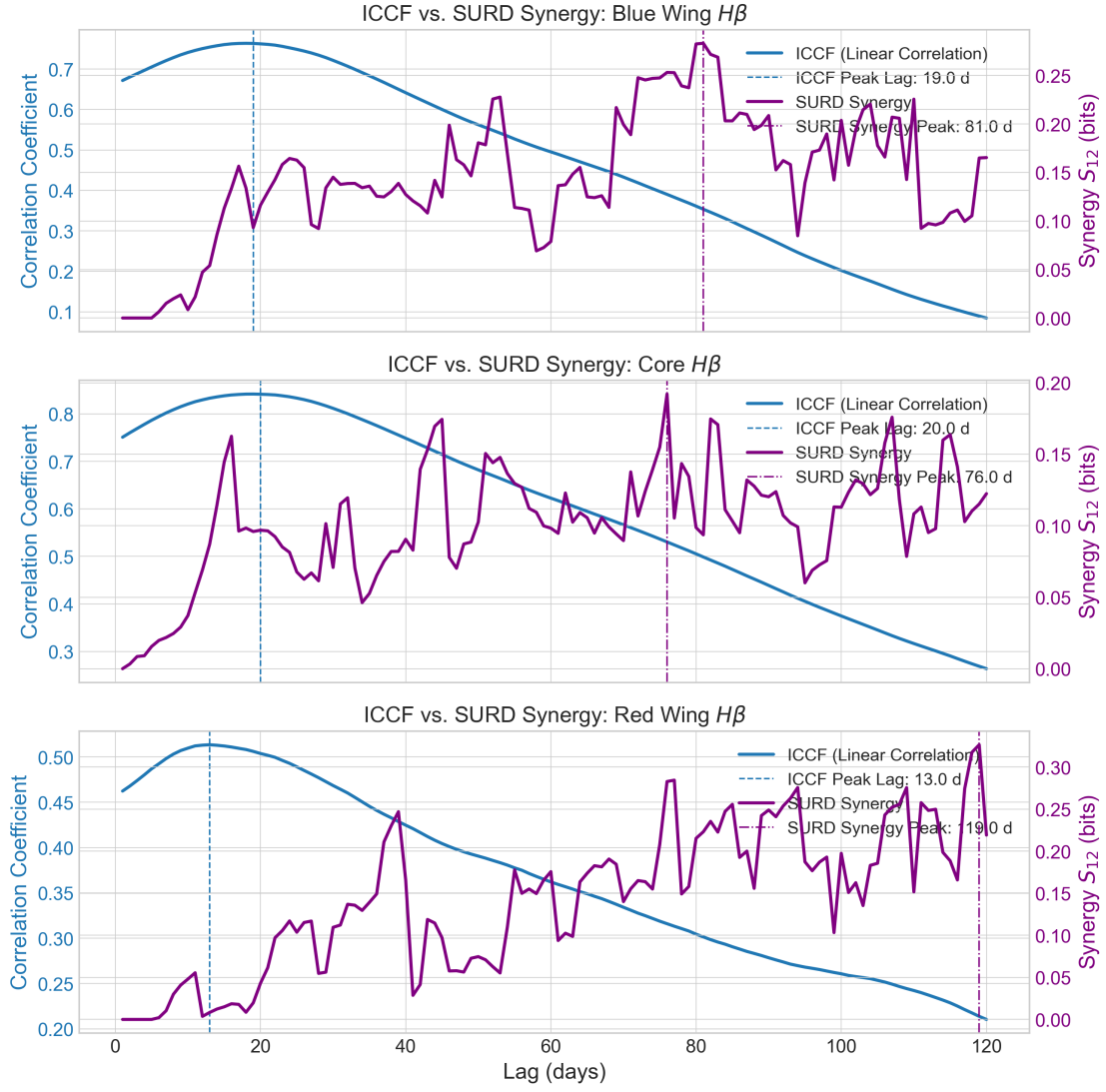


Figure 4: Comparison of the Inter-Correlation Function (ICCF, blue) and the SURD synergy curve (purple) for the Blue Wing (top), Core (middle), and Red Wing (bottom) targets. Vertical lines mark the respective peak lags.

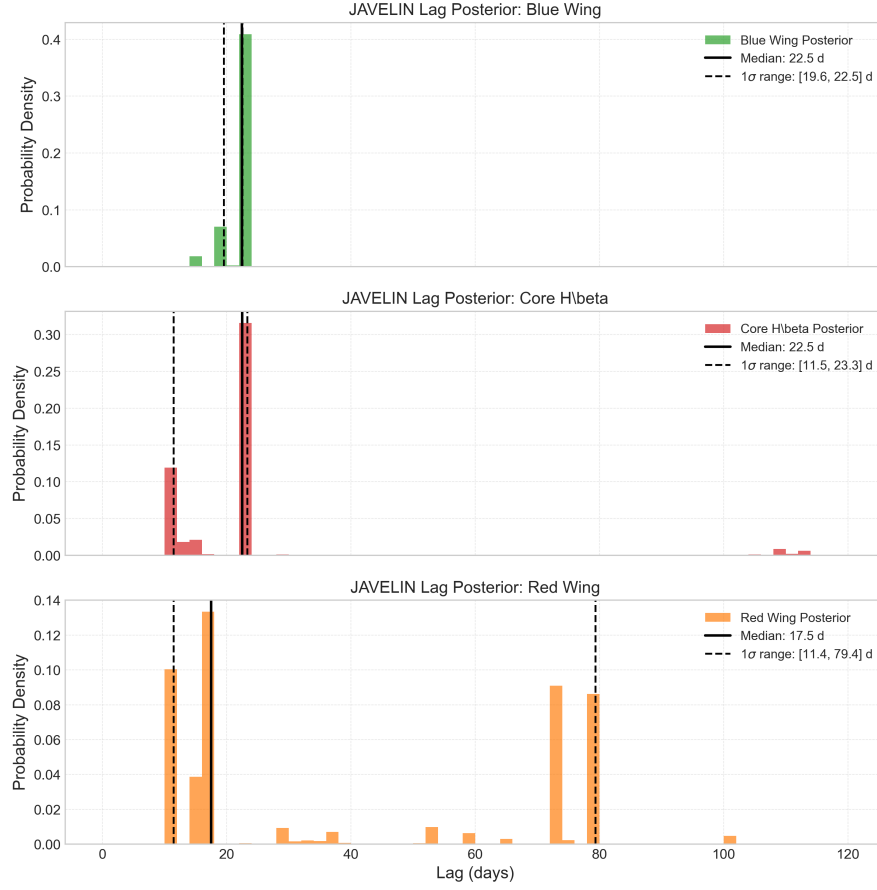


Figure 5: JAVELIN MCMC lag posterior distributions for the Blue Wing (top), Core (middle), and Red Wing (bottom) of $H\beta$ in NGC 5548. Black solid and dashed lines represent the median and 1σ (16th–84th percentiles) confidence intervals, respectively.

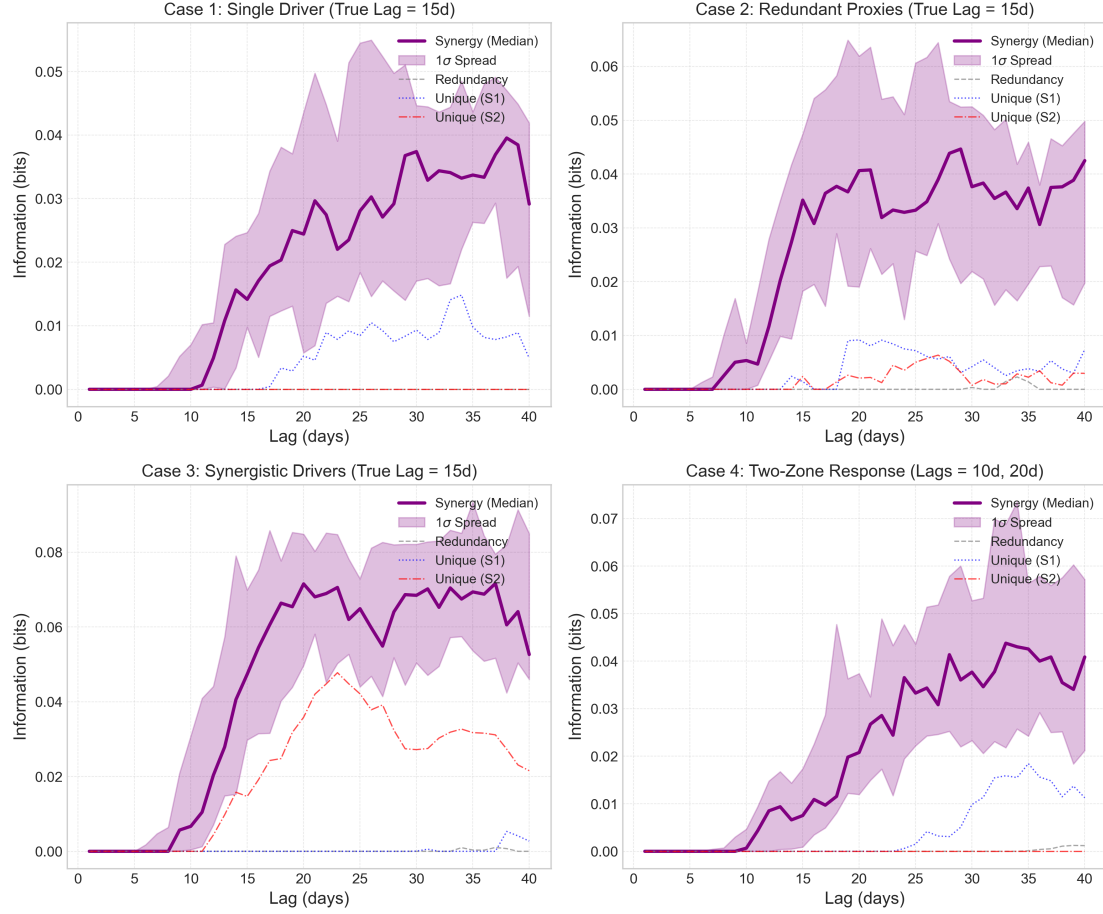


Figure 6: SURD information decomposition scans for the four synthetic benchmark cases run under realistic NGC 5548 observational conditions (real sampling, seasonal gaps, and flux errors). Curves represent the Monte Carlo median Synergy (purple, with 1σ shaded envelope), Redundancy (gray dashed), and Unique information for drivers S1 (blue dotted) and S2 (red dash-dot).

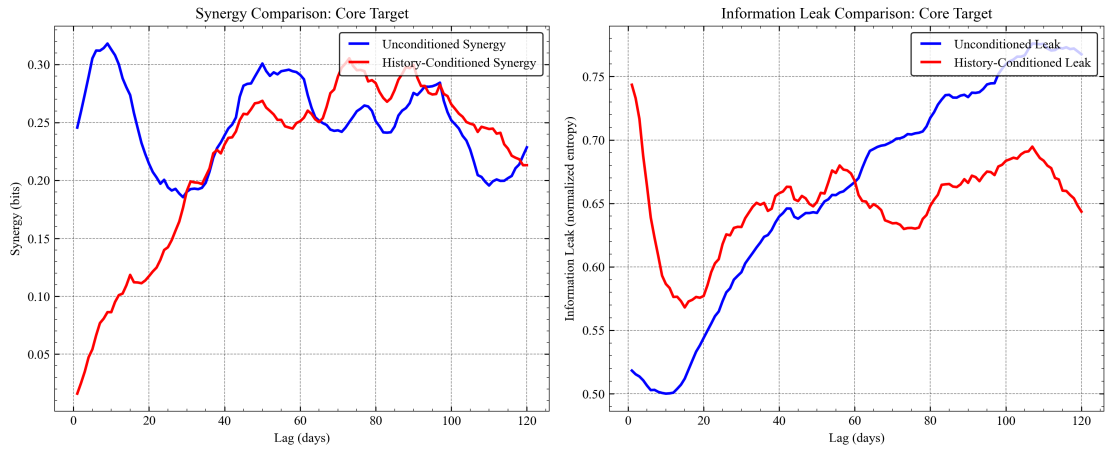


Figure 7: Comparison of the unconditioned (blue) and target-history conditioned (red) SURD synergy (left) and information leak (right) curves for the Core $H\beta$ target. Conditioning on the target's own past shifts the synergy peak from a short autocorrelation-dominated delay of 9 days to a robust, long-term cross-component delay of 73 days.